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Do Language Models Need Music Supervision? Verifiable Rewards for Multi-Constraint Symbolic Music Generation

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Abstract

Language models now generate symbolic music from text, and research has focused on musicality. However, many applications require a score that meets explicit constraints, which models struggle to satisfy jointly: on MusicConstraintBench, our benchmark of 2,180 items over eight families of programmatically verifiable constraints, Llama-3.1-70B satisfies 0.630 of single-constraint items but only 0.044 of four-constraint ones. As a remedy, we introduce MusicRLVR, which trains a language model with group relative policy optimisation (GRPO) on verifier rewards alone, needing no human annotation, reward model or music-domain supervised fine-tuning. MusicRLVR incorporates (1) a hard validation gate that rejects malformed scores, (2) graded per-family credit that, unlike a binary reward, separates partially correct outputs, and (3) an all-satisfied bonus for meeting every constraint at once. Extensive experiments show that, in under four hours of training, MusicRLVR raises Qwen3-4B-Instruct-2507 from 0.160 to 0.797 on mixed constraints, outperforming Llama-3.1-70B, and generalises to unseen property combinations, out-of-range parameters and more constraints than any training prompt. The recipe transfers to Qwen3-8B, and neither trained model loses significant accuracy on general benchmarks.

1 Introduction

Language models can now generate machine-readable symbolic scores from free-text descriptions (Yuan et al., 2024; Qu et al., 2025; Wang et al., 2025a). For users who want a piece that satisfies explicit requirements rather than an arbitrary piece, however, a more natural interface is a score specification that names properties such as key, meter, length, pitch range, final note, rhythmic vocabulary, melodic motion, and form. Unlike perceptual quality (Cideron et al., 2024; Jonason et al., 2025; Ziv et al., 2026), these properties can be evaluated programmatically from the parsed score, placing constraint-following music generation in a class of *property-verifiable generation* problems.

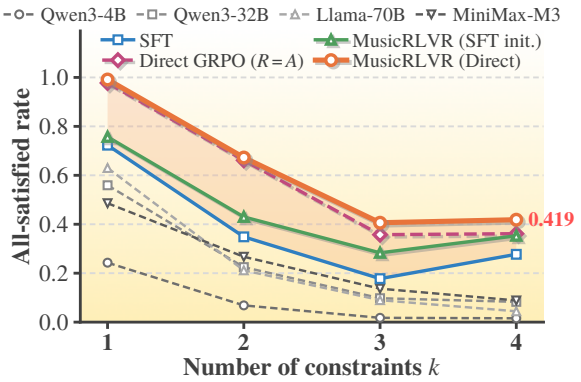


Figure 1: All-satisfied rate against k over all 2,180 MusicConstraintBench items, for the matched runs and the baselines of Table 2. Each slice pools every suite with that k (354, 614, 512 and 700 items), so slices differ in suite and family mixture (Table 51). The right-hand label gives the Direct MusicRLVR rate at $k = 4$.

Despite this progress, current models have a critical drawback: they struggle to satisfy such constraints jointly. A request may simultaneously require a tune to be *in D minor, in* $6/8$*, sixteen bars long, ending on the tonic, with limited melodic leaps and an AABA form*; because these requirements must hold jointly, violating any one of them makes the output non-compliant. No dedicated benchmark isolates this capability, so we construct MusicConstraintBench, a benchmark of 2,180 ABC-generation items spanning eight constraint families. As illustrated in Figure 1, the all-satisfied rate drops sharply as the number of simultaneous constraints increases: Llama-3.1-70B satisfies 0.630 of single-constraint items but only 0.044 of four-constraint ones. Notably, the natural solution of using the deterministic verifiers directly as reinforcement-learning rewards inherits the same conjunction problem when the reward is binary: once any constraint fails, outputs with very different degrees of partial compliance receive the

same reward. In group relative policy optimisation (GRPO), a group whose rollouts all receive the same reward provides no learning signal, and under this binary reward 0.550 of rollout groups in our matched run have zero reward variance over the first 50 updates.

To tackle the above problems, we address the following core question:

Can deterministic verifiers alone teach a model to jointly follow multiple musical constraints?

Hence, we introduce MusicRLVR, a verifier-driven reinforcement learning method that addresses the aforementioned question through three components: (1) a hard validation gate that rejects malformed or unparseable scores before any credit is assigned, since partial credit alone would reward a fragment that satisfies local properties without forming a tune; (2) graded per-family credit that, unlike the binary reward, separates partially correct outputs; and (3) an all-satisfied bonus for meeting every requested constraint at once. We summarise the contributions of this paper as follows:

- **The first benchmark for joint score-constraint compliance.** We introduce MusicConstraintBench, the first benchmark of its kind: 2,180 items spanning eight verifier-defined constraint families, plus 400 paraphrases, testing constraint density, unseen family combinations, parameter extrapolation, and wording robustness.
- **Turning property verifiers into fine-grained training rewards.** We introduce MusicRLVR, which combines a hard validation gate, graded per-family credit, and an all-satisfied bonus without human annotation or a learned reward model, giving credit across requested families and, for form, within one family. Removing or reweighting any of these components lowers the all-satisfied rate over all 2,180 items from 0.580 to 0.502–0.553 (§5.5).
- **Training constraint following without music-domain supervised fine-tuning (SFT).** Experiments show that, in under four hours of training on three accelerators, Direct MusicRLVR raises the Mixed all-satisfied rate of Qwen3-4B-Instruct-2507 from 0.160 to 0.797, outperforming every zero-shot baseline of Table 2, including Llama-3.1-70B, and generalises to property combinations never observed together during training, to parameter values outside the training range and to free-form requests. The recipe transfers to Qwen3-8B, which rises from 0.123 to 0.657 on Mixed, and neither policy loses significant accuracy on IFBench, MMLU-Redux or GSM8K.

2 Related Work

Symbolic music models (Sturm et al., 2016; Huang et al., 2018; Yuan et al., 2024; Qu et al., 2025; Wang et al., 2025a; von Rütte et al., 2023; Thickstun et al., 2023; Agostinelli et al., 2023; Copet et al., 2023) are trained by likelihood or against a learned music–text encoder (Wu et al., 2025), with hard rules applied, if at all, at decoding time (Lu et al., 2021; Huang et al., 2024; Kaliakatsos-Papakostas et al., 2025). Reinforcement learning for music optimises human or learned quality signals (Cideron et al., 2024; Jonason et al., 2025; Ziv et al., 2026; He et al., 2026) or music-theory heuristics (Jaques et al., 2017). However, none of them checks whether the properties a prompt requests hold; in this paper, we instead reward with deterministic verifiers whose families and parameters change from item to item.

Reinforcement learning with verifiable rewards (RLVR) builds on PPO-based RLHF (Schulman et al., 2017; Ouyang et al., 2022) and group-relative baselines (Ahmadian et al., 2024; Shao et al., 2024), underpins reasoning models (Guo et al., 2025; Lambert et al., 2024) and extends beyond mathematics (Ma et al., 2026; Gunjal et al., 2026); DAPO (Yu et al., 2026) and Dr. GRPO (Liu et al., 2025) patch dead groups and length bias, whereas we grade the reward itself. FollowBench (Jiang et al., 2024), IFEval (Zhou et al., 2023) and IFBench (Pyatkin et al., 2026), which also trains with RLVR, test text constraint following, and Wang et al. (2025b) build verifiable sheet-music reasoning questions. However, our constraints act on a parsed score that a fragment can partly satisfy without forming a tune, and no prompt has a target composition.

3 Method

In this section, we introduce MusicRLVR, a reinforcement learning method that trains a general-purpose model to follow requested constraints, with deterministic verifiers as its only task-specific supervision. As shown in Figure 2, this is achieved in four steps: (A) the policy samples G ABC rollouts; (B) a hard validation gate rejects invalid ABC and verifiers check the requested properties (§3.2); (C) a graded reward with an all-satisfied bonus scores the verdicts (§3.3); and (D) GRPO updates the policy with group-relative advantages (§3.4; Algorithm 1).

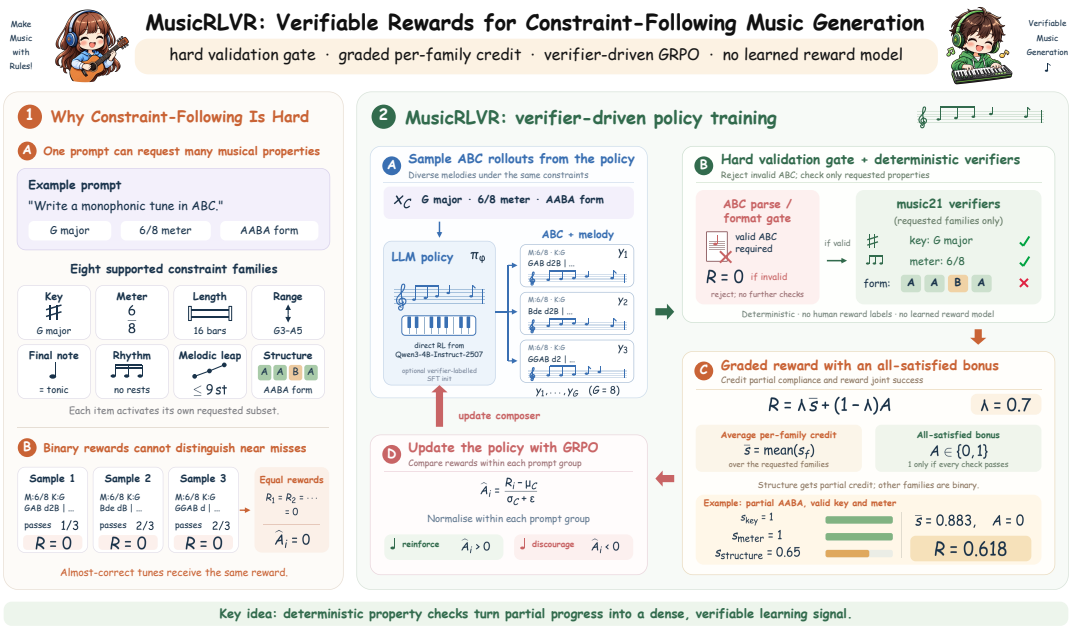


Figure 2: **Overview of MusicRLVR. Left:** Binary all-satisfied rewards cannot distinguish near misses. **Right:** MusicRLVR samples G ABC rollouts, uses a hard validation gate, and verifies requested properties deterministically. Average per-family credit (binary except for graded structure) and an all-satisfied bonus form the reward, normalised per prompt group for GRPO.

3.1 Problem Formulation

In this section, we formalise the task and its success criterion. Let $\mathcal{F}$ be a set of *constraint families*; we use the eight families in Table 1. A *constraint* $c = (f, \theta)$ pairs a family $f \in \mathcal{F}$ with parameters θ (e.g. $f = \texttt{key}$, $\theta = (\mathrm{G}, \mathrm{major})$). Each family has a deterministic *verifier* $v_f : \mathcal{S} \times \Theta_f \to \{0, 1\}$, where $\mathcal{S}$ is the set of parseable scores and Θ_f the parameter space of family f. A *problem* is a set of constraints $C = \{c_1, \ldots, c_k\}$ with distinct families, rendered into a prompt x_C by a fixed template. The policy π_ϕ, with trainable parameters ϕ, emits a string y in ABC text notation (Appendices C.5–C.7); ρ extracts the ABC and applies the validation gate of Appendix C.1, returning $\rho(y) \in \mathcal{S}$ or $\perp$ on rejection. The training reward uses the binary indicator

$$A(y, C) = \begin{cases} \prod_{(f,\theta)\in C} v_f(\rho(y), \theta) & \rho(y) \neq \perp, \\ 0 & \rho(y) = \perp. \end{cases} \tag{1}$$

At evaluation, all-satisfied rate is the fraction of all test items whose outputs pass the validation gate and satisfy every requested musical constraint, i.e. the mean of A.

3.2 Constraint Families and Verifiers

In this section, we present the verifiers that supply every reward signal, how they are validated and how the key criterion is calibrated.

Verifying properties at multiple musical scales. Table 1 lists the eight constraint families. Seven constrain one attribute each, such as key, meter or pitch range; the eighth, `structure`, is a cross-section form constraint: a sixteen-bar tune in a stated form over four-bar units (AABA, AABB, ABAB), units sharing a letter identical note-for-note and units with different letters differing. A unit u_a ($a \in \{1, \ldots, 4\}$) is its sequence of $M = 4$ bar signatures (pitch and duration per event); a tune without exactly $4M$ bars scores 0 on every structure term. Writing E for the index pairs required equal, D for those required different (minimal sets per form in Appendix C.1) and $\mathbb{I}[\cdot]$ for an indicator,

$$v_{\texttt{structure}} = \mathbb{I}\Big[\textstyle\bigwedge_{(a,b)\in E} u_a = u_b\Big] \cdot \mathbb{I}\Big[\textstyle\bigwedge_{(a,b)\in D} u_a \neq u_b\Big]. \tag{2}$$

We write v_{rep} and v_{contrast} for the two factors, the *repetition* and *contrast* halves. Their conjunction is zero whenever either fails, however close the output is. Writing $u_a[m]$ for the m-th bar signature of unit u_a, the graded repetition credit

$$\bar{e} = \frac{1}{|E|M} \sum_{(a,b)\in E} \sum_{m=1}^{M} \mathbb{I}\big[u_a[m] = u_b[m]\big] \tag{3}$$

counts how much of the required repetition is already in place rather than whether all of it is. The verifiers are validated without annotators: all 32

Table 1: The eight constraint families, each decided by a deterministic `music21` routine on the parsed score, spanning global, bar, note, transition and cross-section properties. The third column is the default sampler's domain; the verifier domain Θ_f is wider, which the Edge split exploits. Table 20 adds each prompt sentence.

Family	Checked over	Default sampler values
`key`	global	13 keys: 8 major, 5 minor
`meter`	every bar	$4/4$, $3/4$, $6/8$, $2/4$; each bar must sum to the signature
`length`	global	8, 12 or 16 bars, no pickup
`range`	every note	4 inclusive pitch spans, e.g. G3–A5
`final`	last note	a pitch class, any octave (a scale degree 1/3/5 when a key is present)
`rhythm`	every note	4 admissible duration sets; no dotted values, no rests
`leap`	adjacent notes	largest melodic interval: 7, 9 or 12 semitones
`structure`	4-bar units	AABA, AABB or ABAB over 16 bars; equal letters identical, different letters distinct

constructed scores with known property values, covering the eight families, receive the correct verdict; six single-property mutations of a fully compliant tune each flip exactly the targeted family; and the meter check agrees with the declared meter of all 400 IrishMAN tunes tested (Appendix C.1).

Calibrating the key criterion. The `key` verifier is the one hybrid among the eight, because a literal check fails in both directions: the `K:` header is gameable, and key-profile analysis (Krumhansl, 2001; Temperley, 1999) (Aarden–Essen profiles) agrees with the corpus header on only 0.750 of 400 corpus tunes. We accept a tune iff the header matches *and* the analysed key is the requested one or its relative (Appendix C.3).

3.3 Gated Graded Reward

In this section, we present the reward that turns verifier verdicts into a learning signal.

Binary all-satisfied reward and its limitations. For a prompt x_C, GRPO samples a group of G rollouts $\{y_1, \ldots, y_G\}$ and centres their rewards within that group,

$$\hat{A}_i = \frac{R(y_i, C) - \mu_C}{\sigma_C + \epsilon}, \quad \mu_C = \frac{1}{G}\sum_{j=1}^{G} R(y_j, C),$$
$$\sigma_C^2 = \frac{1}{G-1}\sum_{j=1}^{G}\big(R(y_j, C) - \mu_C\big)^2, \tag{4}$$

with $\epsilon = 10^{-4}$ a small constant. Despite matching the evaluation criterion exactly, taking $R = A$ (Equation 1) has two limitations: (1) it separates no partial ordering, as a rollout that misses one constraint scores the same as one that misses all; and (2) such ties remove the learning signal, since a group whose rollouts all score alike contributes only the Kullback–Leibler (KL) term (Appendix D.24).

Addressing the limitations by graded per-family credit. To address the aforementioned limitations, we grade the reward per requested family and keep an all-satisfied bonus. In detail, for a completion y on problem C with $k = |C|$,

$$R(y, C) = \begin{cases} 0, & \rho(y) = \bot, \\ \lambda \dfrac{1}{k} \displaystyle\sum_{(f,\theta)\in C} s_f(\rho(y), \theta) & \\ \quad + (1-\lambda)\, A(y, C), & \text{else}, \end{cases} \tag{5}$$

with $\lambda = 0.7$, which leaves the binary term 0.3 of the reward even at $k = 1$ (§5.5 and Appendix D.18 vary it). Here $s_f : \mathcal{S} \times \Theta_f \to [0, 1]$ equals v_f for the seven non-structure families; for `structure` it is graded, $s_{\texttt{structure}} = 0.7\,\bar{e} + 0.3\,v_{\text{contrast}}$ (a fixed split, separate from λ). The λ-weighted average separates rollouts satisfying different *numbers of families*; holding the contrast verdict fixed, the graded $s_{\texttt{structure}}$ also separates rollouts that fail the form check but match different fractions of bar positions, which is credit *within* a family. We grade repetition and keep contrast binary because exact bar matches measure partial repetition directly (a symmetric variant is in Appendix E.2).

Guarding against fragments by a hard validation gate. However, paying partial credit to every output would let a fragment satisfying `key` and `range`, but forming no tune, collect $\lambda \cdot 2/k$ (cf. Skalse et al., 2022). As a remedy, the gate scores every rejected output 0 before any credit is assigned. Training and main evaluation share the gate; under the stricter format checks of Appendix C.2, the matched checkpoints of Table 2 keep every success.

3.4 Training

In this section, we elaborate on the training procedure, which consists of two parts, namely (1) GRPO on the reward of Equation 5; and (2) an

optional music-domain SFT warm start. *Direct* applies (1) to Qwen3-4B-Instruct-2507, whereas *SFT init.* applies (2) before (1); both use the same reward, prompt pool and GRPO settings, with the initial policy as the KL reference.

Optimising with the verifier as the only source of task supervision. In detail, both routes use GRPO (Shao et al., 2024): every rollout of a group is first scored by Equation 5, its group-relative advantage (Equation 4) is then assigned to every token, and the policy finally maximises the PPO clipped surrogate (Schulman et al., 2017) with a per-token KL penalty β towards the route's initial checkpoint (Equation 6 in Appendix A). No value network is trained; the group mean μ_C is the baseline. We run $G = 8$, $\beta = 0.04$, clipping range $\varepsilon_{\text{clip}} = 0.2$, learning rate 1×10^{-6}, 300 optimiser steps on 6,000 synthetic prompts, with Qwen3-4B-Instruct-2507 (Yang et al., 2025) trained full-parameter (§B.1). The prompt pool upsamples structure-containing problems from 21% to 33% (§5.5 compares both), and Table 7 gives all settings.

Warm-starting on targets that the verifiers label themselves. The optional SFT uses 12,890 instruction–tune pairs self-labelled from the IrishMAN corpus (Wu et al., 2023): for each corpus tune the verifiers read candidate constraint parameters off the parsed score, sample $k \sim \mathcal{U}\{1, 2, 3\}$ families, and admit the pair only if $A(y, C(y)) = 1$. Held-out family pairs (§4) are excluded from both training pools. Full construction details are in Appendices B.4–B.6.

4 MusicConstraintBench

In this section, we introduce MusicConstraintBench, which comprises five complementary tests: *constraint density* (Single, Mixed, High-k), *compositional generalisation* (Seen, Unseen, Dense), *parameter extrapolation* (Edge), *multi-section control* (Structure), and *linguistic variation* (paraphrases). Single ($n = 200$) contains one constraint, High-k ($n = 160$) four, and Mixed ($n = 300$) spans $k \in \{1, 2, 3, 4\}$ and serves as the per-family diagnostic. Seen ($n = 200$) contains two or three families that co-occur in training, whereas Unseen ($n = 200$) requires one of four held-out pairs; Structure ($n = 120$) always includes the cross-section form constraint. Dense ($n = 500$; four constraints with a held-out pair) and Edge ($n = 500$; parameters outside the RL prompt domain, §5.4) complete

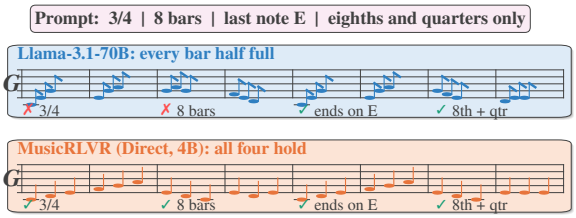


Figure 3: One High-k item, the first that Direct MusicRLVR solves and Qwen3-32B, Llama-3.1-70B and MiniMax-M3 all fail. Llama-3.1-70B fills half of every 3/4 bar.

the 2,180 items; both were added after the reward design was final. The held-out pairs are (`key`, `rhythm`), (`meter`, `leap`), (`final`, `structure`), and (`length`, `range`); none occurs together in either training stage (Lake and Baroni, 2018; Kim and Linzen, 2020). They form a perfect matching over the eight families, so each family is held out once and any held-out-pair-free set has at most $k = 4$ (Appendix C.4). Finally, re-rendering all Single and Seen items with alternative wording yields a 400-item paraphrase suite with unchanged constraint values (Appendix C). Figure 3 shows one High-k item that Llama-3.1-70B fails and Direct MusicRLVR solves.

5 Experiments

In this section, we carry out experiments to address the following questions:

- **Q1**: Do existing models comply? See §5.2.
- **Q2**: Is music-domain SFT needed? See §5.3.
- **Q3**: Does the gain generalise? See §5.4.
- **Q4**: Which reward parts matter? See §5.5.
- **Q5**: Where are the gains largest? See §5.6.

Additional results can be found in Appendix D.

5.1 Experiment Setup

To answer the above five questions, we score greedy completions with the all-satisfied criterion of §3 (temperature 1.0 for the sampling-budget study). **Baselines.** Qwen3-4B-Instruct-2507, Qwen3-8B and Qwen3-32B (Yang et al., 2025), Llama-3.1-70B (Grattafiori et al., 2024), MiniMax-M3, and ChatMusician-7B (Yuan et al., 2024) all receive the same user prompt under their own chat templates without requested reasoning; the music-domain SFT checkpoint serves as a supervised baseline (Appendix B.5). **Training runs.** The three *matched runs*, Direct GRPO ($R = A$, i.e. $\lambda = 0$), Direct MusicRLVR, and SFT-initialised MusicRLVR, share seed 43, 300 updates, and 48 completions per update, and are evaluated once per suite (decoding variation: Appendix B.10; paired McNemar tests

Table 2: All-satisfied rate under greedy decoding on the eight MusicConstraintBench suites and over all 2,180 items. Qwen3-4B is Qwen3-4B-Instruct-2507; the 4B RL rows are matched runs at equal budget, and RFT (rejection-sampling fine-tuning) fine-tunes Qwen3-4B on its own verifier-accepted samples at that budget (Appendix D.7). Each column marks its best and runner-up; the shaded row is our main configuration. Counts and paired tests: Appendix D.13.

	Six original splits						Later splits		Overall
Model	Mixed	Single	Seen	Unseen	High-k	Structure	Dense	Edge	All
Zero-shot									
Qwen3-4B	0.160	0.225	0.100	0.040	0.037	0.100	0.008	0.010	0.068
Qwen3-8B	0.123	0.210	0.115	0.050	0.006	0.000	0.020	0.032	0.064
Qwen3-32B	0.363	0.535	0.250	0.190	0.094	0.267	0.074	0.112	0.204
Llama-3.1-70B	0.380	0.695	0.300	0.195	0.056	0.167	0.040	0.058	0.197
MiniMax-M3	0.367	0.580	0.315	0.190	0.100	0.008	0.088	0.160	0.215
ChatMusician-7B	0.160	0.290	0.130	0.090	0.006	0.000	0.020	0.046	0.084
Trained (Qwen3-4B-Instruct-2507)									
SFT	0.477	0.745	0.415	0.440	0.231	0.600	0.290	0.076	0.346
RFT	0.543	0.960	0.445	0.355	0.163	0.758	0.098	0.128	0.342
Direct GRPO ($R = A$)	0.797	0.960	0.745	0.565	0.469	0.908	0.322	0.300	0.545
MusicRLVR (SFT init.)	0.577	0.880	0.585	0.530	0.350	0.383	0.366	0.134	0.424
MusicRLVR (Direct)	**0.807**	**0.985**	**0.815**	0.595	**0.575**	**0.967**	0.362	**0.310**	**0.580**
Trained (Qwen3-8B)									
MusicRLVR (Direct)	0.657	0.915	0.705	**0.645**	0.431	0.900	**0.404**	0.252	0.530

(McNemar, 1947): Appendix D.13). Reward ablations (§5.5) start from the Direct configuration; sensitivity analyses use a separate SFT-initialised reference, the *diagnostic study*.

5.2 A1: Existing Models Struggle with Multi-Constraint Music Instructions

In this section, we evaluate the zero-shot models on MusicConstraintBench. As shown in Table 2 and Figure 1, we observe that (1) parsing is largely solved while joint compliance remains low: Llama-3.1-70B writes a parseable score for 0.970 of Mixed items yet satisfies every requested constraint on only 0.380, with Qwen3-32B and MiniMax-M3 alongside it at 0.363 and 0.367; (2) neither size nor music specialisation sets the ranking: Qwen3-8B reaches 0.123 on Mixed, below the 0.160 of the smaller Qwen3-4B, and the music-specialised ChatMusician-7B reaches only 0.160; (3) compliance is lost as constraints accumulate: Llama-3.1-70B falls 14.2× from one to four constraints, from 0.630 to 0.044, whereas Direct MusicRLVR falls 2.4×, from 0.992 to 0.419 (Appendix D.20).

5.3 A2: MusicRLVR Teaches Constraint Following Without Music-Domain SFT

In this section, we train Qwen3-4B-Instruct-2507 with MusicRLVR directly, without music-domain SFT. As shown in Table 2, we observe that (1) Direct MusicRLVR raises the Mixed all-satisfied rate from 0.160 to 0.797 and surpasses every zero-shot baseline of Table 2 on all eight suites, reaching 0.985 on Single, 0.967 on Structure and 0.575 on High-k against a best zero-shot of merely 0.100,

Table 3: One greedy decode against the best of eight samples reranked by the verifier (temperature 1.0; Appendix D.8). Each row marks its best and runner-up.

	Qwen3-4B (zero-shot)		MusicRLVR (Direct)	
Suite	Greedy	Best of 8	Greedy	Best of 8
Mixed	0.160	0.393	0.807	**0.897**
Single	0.225	0.685	0.985	**1.000**
Seen	0.100	0.330	0.815	**0.875**
Unseen	0.040	0.180	0.595	**0.750**
High-k	0.037	0.050	0.575	**0.756**
Structure	0.100	0.367	0.967	**0.975**
Dense	0.008	0.052	0.362	**0.624**
Edge	0.010	0.090	0.310	**0.448**

with every completion clearing the validation gate on the six original suites. Item by item, it satisfies 609 of the 2,180 items that Qwen3-32B, Llama-3.1-70B and MiniMax-M3 all fail, against 5 the other way round (Appendix F). (2) Music-domain SFT is not a prerequisite: under the same reward, prompt pool and budget, the SFT-initialised route reaches 0.577 on Mixed against 0.797 for Direct and trails on seven of eight suites; pooled over the 2,180 items they disagree on 765, 549 in Direct's favour ($p < 10^{-4}$), and Direct stays significantly ahead on six of the eight suites after Holm correction (Appendix D.13). (3) The supervised initialisation also collapses on form, clearing the repetition half of the form verifier on 119/120 Structure items but the contrast half on only 50, and emitting an all-identical tune on 69 (Figure 4(f)).

Neither self-distillation nor inference-time selection substitutes for training. We further observe that (1) fine-tuning Qwen3-4B on its own verifier-accepted samples, drawn at the sampling budget of GRPO, reaches 0.543 on Mixed and trails Direct on

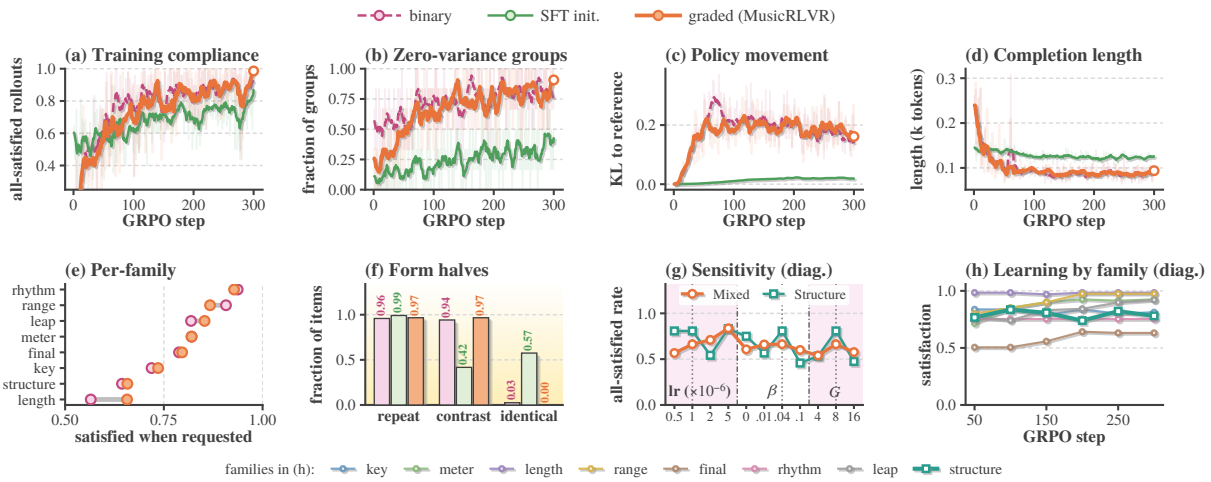


Figure 4: **Training dynamics.** (a–f) Matched runs of Table 2: "binary" and "graded" are the two Direct arms ($\lambda = 0$ and 0.7), "SFT init." starts from the SFT checkpoint. **(a)** All-satisfied rollouts. **(b)** Zero-variance groups. **(c)** KL. **(d)** Completion length. **(e)** Per-family satisfaction. **(f)** Form-verifier components. (g–h) Diagnostic study: **(g)** sensitivity to learning rate, β and G (dotted: reference); **(h)** per-family learning curves.

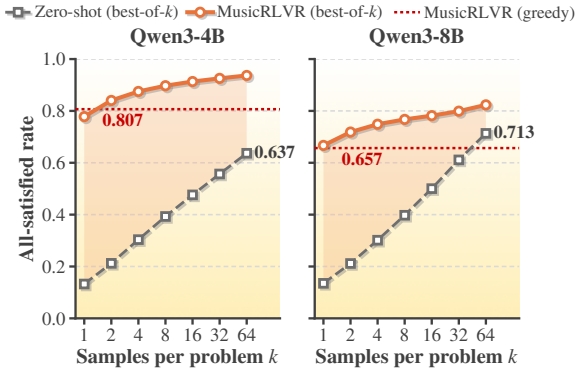


Figure 5: Best-of-k verifier reranking on Mixed (64 samples at temperature 1.0). The dotted line is one greedy decode of the trained policy.

Table 4: Accuracy (%) before and after training on IFBench (Pyatkin et al., 2026), MMLU-Redux (Gema et al., 2025) and GSM8K (Cobbe et al., 2021). No difference is significant (paired exact McNemar, $p \geq 0.096$). Shading as in Table 2.

	IFBench			
Model	strict	loose	MMLU-Redux	GSM8K
Qwen3-4B (zero-shot)	32.0	33.3	83.9	93.9
MusicRLVR (Direct)	29.3	31.7	84.4	94.1
Qwen3-8B (zero-shot)	25.7	30.3	81.5	93.0
MusicRLVR (Direct, Qwen3-8B)	25.3	30.0	82.6	93.3

all eight suites (Table 2; Appendix D.7); (2) reranking 64 samples of the untrained Qwen3-4B with the verifier reaches 0.637, below Direct's single greedy decode, and the untrained Qwen3-8B needs 32 to 64 reranked samples to match its trained policy (Figure 5); (3) reranking eight samples of the trained policy raises every suite further (Table 3), to 0.897 on Mixed against 0.393 for the untrained model, so training and selection are complementary.

The recipe also carries over to Qwen3-8B (Appendix B.3), which rises from 0.123 to 0.657 on

Table 5: All-satisfied rate beyond the training range of k (at most 3). $k = 4$ is Dense; $k = 5$–7 hold 300 items each, every one with a held-out pair. Colours as in Table 2.

Model	$k = 4$	$k = 5$	$k = 6$	$k = 7$
Qwen3-4B (zero-shot)	0.008	0.003	0.000	0.000
SFT	0.290	0.140	0.117	0.073
RFT	0.098	0.053	0.007	0.003
Direct GRPO ($R = A$)	0.322	0.170	0.060	0.063
MusicRLVR (SFT init.)	0.366	0.187	0.200	0.117
MusicRLVR (Direct)	0.362	0.230	0.160	0.087
MusicRLVR (Direct, Qwen3-8B)	**0.404**	**0.273**	**0.250**	**0.153**

Mixed and posts the highest Unseen and Dense rates in Table 2. As shown in Table 4, neither policy trades away general capability (Appendix D.4).

5.4 A3: MusicRLVR Generalises Beyond the Training Distribution

In this section, we test the trained policy beyond its training distribution. As shown in Tables 2 and 5, we observe that (1) the gain transfers to held-out pairs. Every Unseen item requires one of the four family pairs excluded from both training pools; Direct reaches 0.595 there, three times the best zero-shot rate of 0.195 (Appendix B.8).

(2) It extends to new parameter values. Edge moves meters to 5/4, 2/2, 9/8, 12/8 and lengths to 4, 24, 32 bars and extends leap, range and rhythm (Table 14). Direct scores 0.300 against the strongest zero-shot 0.160. On items whose meter or length no RL prompt requests, Direct reaches 0.287 and 0.181, against at most 0.138 and 0.142 for any zero-shot model (Table 33).

(3) It extends to more constraints than trained. Training prompts request at most three constraints (Appendix B.7), so every High-k and Dense item already asks for one more; Direct reaches 0.575

Table 6: Reward ablations in the Direct configuration. Each row below the default changes one part of Equation 5, keeping the seed, budget, prompts (except the last row) and evaluation of Table 2; $\lambda = 0$ is Direct GRPO ($R = A$). Details: Appendix D.10. Colours as in Table 2.

Variant	Mixed	Single	Seen	Unseen	High-k	Structure	Dense	Edge	All
MusicRLVR (Direct)	**0.807**	**0.985**	**0.815**	0.595	**0.575**	**0.967**	**0.362**	**0.310**	**0.580**
Gate on A only	0.763	0.975	0.770	0.575	0.469	0.867	0.346	0.238	0.534
$\lambda = 0$ ($R = A$)	0.797	0.960	0.745	0.565	0.469	0.908	0.322	0.300	0.545
$\lambda = 0.5$	0.703	**0.985**	0.665	0.545	0.406	0.883	0.246	0.302	0.502
$\lambda = 1$ (no bonus)	0.753	0.950	0.735	**0.635**	0.494	0.908	0.358	0.298	0.553
Binary form credit	0.777	0.980	0.785	0.590	0.481	0.900	0.360	0.242	0.546
Symmetric form credit	0.757	0.950	0.760	0.500	0.394	0.942	0.282	0.272	0.515
Equal form weights	0.753	0.940	0.740	0.570	0.475	0.942	0.352	0.290	0.544
Form share 21%	0.753	0.980	0.740	0.550	0.381	**0.967**	0.270	0.276	0.518

and 0.362 there. Suites at $k = 5$, 6 and 7 go further (Appendix D.9): the untrained Qwen3-4B satisfies 1 of their 900 items, whereas Direct MusicRLVR keeps 0.230, 0.160 and 0.087, and the Qwen3-8B policy 0.273, 0.250 and 0.153, the highest at every rung (Table 5).

(4) It extends to free-form wording. On 60 hand-written requests that phrase the constraints as a user might (Appendix D.3), Direct reaches 0.583, the highest of the 4B models (Qwen3-4B: 0.067), and the Qwen3-8B policy reaches 0.717.

5.5 A4: Every Reward Component Helps

In this section, we ablate Equation 5 in the Direct configuration, one part at a time. As shown in Table 6, every change lowers the rate over all 2,180 items from 0.580 to 0.502–0.553 (paired exact McNemar, Holm-adjusted $p \leq 0.002$). The default is higher on 61 of the 64 variant–suite pairs, and every variant loses most on High-k, 0.381–0.494 against 0.575, where at least four families must hold at once. We further observe that (1) crediting outputs that the gate rejects but music21 parses lowers all eight suites, Structure by 0.100 and High-k by 0.106; (2) dropping per-family credit ($\lambda = 0$) or the all-satisfied bonus ($\lambda = 1$) lowers High-k to 0.469 and 0.494; (3) graded form credit lifts Structure to 0.967 from 0.900–0.942 under binary, symmetric and equal-weight form rewards, and training with 21% form prompts gives the lowest High-k rate, 0.381, even though that pool holds more three-constraint prompts.

We also observe that (1) graded credit cuts groups without a reward gradient from 0.550 to 0.307 over the first 50 updates and from 0.717 to 0.660 over all 300 (Figure 4(b)); (2) in the matched runs, with each family scored only on the items requesting it, gains are clearly largest for `length` (0.558 → 0.663) and `leap` (0.833 → 0.839), while the other six move in absolute value by at most 0.019 (Figure 4(e); Appendix D.1); (3) from the SFT checkpoint, graded form credit lifts Structure from 0.525 to 0.808 (Appendix D.15), and every learning rate, KL coefficient and group size tested keeps Mixed at 0.540–0.837, above the SFT start of 0.477 (Figure 4(g)).

5.6 A5: The Largest Gains Are on Counting and Structure

In this section, we break the all-satisfied rate down by constraint family over all 2,180 items (Appendix D.1) and observe that (1) larger zero-shot models already handle the tonal families: on `key` and `final`, Qwen3-32B and Llama-3.1-70B reach 0.719–0.868 against 0.242 and 0.278 for Qwen3-4B, and Direct MusicRLVR stays close with 0.735 and 0.796; (2) scale falls short on counting and structure: no zero-shot model exceeds 0.378 on `meter`, 0.281 on `length` or 0.150 on `structure`, whereas Direct reaches 0.820, 0.657 and 0.658; (3) Direct also far exceeds every zero-shot model on `rhythm`, `range` and `leap`, 0.853–0.928 against at most 0.671.

6 Conclusion and Future Work

In this paper, we frame constraint-following music generation as a *property-verifiable generation* problem and introduce MusicConstraintBench and MusicRLVR, a method using only verifier rewards. Our results show that MusicRLVR raises Qwen3-4B from 0.160 to 0.797 on Mixed in under four hours of training, above every zero-shot baseline including Llama-3.1-70B, with gains beyond the training distribution and no significant change on general benchmarks. This demonstrates that music models need no music-domain SFT to follow musical constraints. For future work, we will extend the verifiers to polyphony and richer forms.

Limitations

The verifiers cover eight constraint families over monophonic ABC in Western tonal idioms; polyphony and richer forms are outside this scope. Within it, held-out family pairs, out-of-range parameters, more constraints than any training prompt and hand-written free-form requests test generalisation (§5.4). We evaluate compliance with stated constraints rather than musical quality, which a listening study would measure.

Ethics Statement

The study involves no human subjects, no personal data and no user-facing deployment. Supervised targets derive from the publicly available IrishMAN corpus of traditional Irish music (Wu et al., 2023); the reinforcement prompts are produced by deterministic samplers rather than drawn from a corpus. The intended uses are settings that need a score to meet a stated specification, such as music education, game audio and accessibility tools. The misuse we consider most likely is reading a compliance score as a quality score; the verifiers certify the requested properties. Appendix G gives the longer discussion.

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Contents of Appendix

A Algorithm **12**

B Experimental Settings **12**
- B.1 Infrastructure . . . 12
- B.2 Training Hyperparameters . . . 13
- B.3 Second Base Model . . . 13
- B.4 SFT Data Construction . . . 14
- B.5 SFT Training Examples . . . 15
- B.6 Prompt Template . . . 15
- B.7 Dataset Statistics . . . 15
- B.8 Prompt Coverage, Training Overlap, and Split Composition . . . 15
- B.9 Reproducibility . . . 17
- B.10 Evaluation Nondeterminism . . . 17
- B.11 Data and Code Release . . . 18

C Benchmark and Verifier Details **19**
- C.1 Verifier Implementation Details . . . 19
- C.2 Stricter Fixed-Instruction Gate Audit . . . 20
- C.3 Key Verifier Calibration . . . 21
- C.4 Held-out Pair Rationale . . . 22
- C.5 ABC Notation Primer . . . 22
- C.6 ABC Notation: A Worked Example . . . 22
- C.7 Constraint Rendering Examples . . . 23
- C.8 Paraphrase Examples . . . 23

D Additional Experiments **24**
- D.1 Per-Family Satisfaction Across Models . . . 24
- D.2 Instruction Wording . . . 25
- D.3 Free-Form Requests . . . 25
- D.4 General Capability . . . 26
- D.5 Paraphrase and Training Dynamics . . . 26
- D.6 Rejection Sampling and a Music-Specialised Baseline . . . 26
- D.7 Rejection-Sampling Fine-Tuning . . . 27
- D.8 Training versus Inference-Time Selection . . . 27
- D.9 Constraint-Count Ladder . . . 27
- D.10 Direct Reward Ablations . . . 28
- D.11 Edge Subsets and a Stricter Key Criterion . . . 28
- D.12 Significance Tests . . . 29
- D.13 Matched Comparison . . . 29
- D.14 Diagnostic Reference Results . . . 30
- D.15 Diagnostic Reward-Design Ablations . . . 30
- D.16 Fixed-Checkpoint Evaluation . . . 30
- D.17 Hyper-parameter Sweeps . . . 30
- D.18 Reward Mixing Weight . . . 31
- D.19 Per-Family Satisfaction on the Original Suites . . . 32
- D.20 Performance by Number of Constraints (k) . . . 32
- D.21 Variety and Degeneration of the Matched Checkpoints . . . 32
- D.22 Length-Controlled Output Statistics . . . 35
- D.23 Output Length Statistics . . . 35
- D.24 Zero-Variance Groups During Training . . . 35
- D.25 Additional Training Curves . . . 36
- D.26 Additional Diagnostics . . . 36

E Error and Structure Analysis **36**
- E.1 Which Constraint Binds, by Split . . . 36
- E.2 Structure Reward Variants . . . 36
- E.3 Structure Results by Form Type . . . 38

F Qualitative Examples **39**

G Broader Impact **48**

A Algorithm

In this section, we present the detailed algorithm of MusicRLVR. Algorithm 1 runs verifier-driven GRPO directly or with an optional supervised warm start built from verified corpus-derived labels ($\theta_f(y) := p_f(\rho(y))$, §3.4). Unless marked Direct, the diagnostics below describe the SFT-initialised route, retaining a common initialisation for the reward-design comparisons.

Objective. For a prompt x_C from the reinforcement pool $\mathcal{D}_{\mathrm{RL}}$, the behaviour policy $\pi_{\phi_{\text{old}}}$ samples a group $\{y_1, \ldots, y_G\}$ and the advantage $\hat{A}_i$ of Equation 4 is assigned to every token of y_i. Writing $y_{i,t}$ for the t-th token of y_i and $r_{i,t}(\phi)$ for the token-level likelihood ratio $\pi_\phi(y_{i,t} \mid x_C, y_{i,<t}) / \pi_{\phi_{\text{old}}}(y_{i,t} \mid x_C, y_{i,<t})$, GRPO maximises

$$\mathcal{J}(\phi) = \mathbb{E}\Big[\tfrac{1}{G}\sum_{i=1}^{G}\tfrac{1}{|y_i|}\sum_{t=1}^{|y_i|}\Big\{\min\big(r_{i,t}\hat{A}_i, \operatorname{clip}(r_{i,t}, 1-\varepsilon_{\text{clip}}, 1+\varepsilon_{\text{clip}})\hat{A}_i\big) - \beta\,\widehat{K}_{i,t}\Big\}\Big], \tag{6}$$

where the expectation is over problems C and their sampled groups, $|y_i|$ counts completion tokens, $\varepsilon_{\text{clip}} = 0.2$ is the clipping range, $\widehat{K}_{i,t} = e^d - d - 1$ with $d = \log \pi_{\text{ref}}(y_{i,t} \mid \cdot) - \log \pi_\phi(y_{i,t} \mid \cdot)$ is the per-token KL estimator of Shao et al. (2024), $\pi_{\phi_{\text{old}}}$ is a detached copy of π_ϕ (one update per rollout batch), and π_{ref} is the route's initial checkpoint (hyperparameters in Appendix B.2).

B Experimental Settings

In this section, we present the detailed experimental settings, including the infrastructure, hyperparameters, SFT data, prompt template and dataset statistics.

B.1 Infrastructure

In this section, we describe the training stack, which uses ms-swift's GRPO implementation with a custom reward plugin that shells out to the evaluation verifiers, so training and benchmarking share one `music21` build. On these accelerators, co-locating the vLLM rollout engine with the training state is infeasible for full-parameter 4B training: the sleeping engine's memory pool still counts against the training process and the backward pass fails at a fixed allocation point regardless of batch size. Serving rollouts from a dedicated device solves it. Two further settings are required on this stack:

Algorithm 1: MusicRLVR: verifier-only GRPO with optional self-labelled SFT

Input: initial policy π_{ϕ_0}; optional SFT flag u and corpus tunes $\mathcal{Y}$; families $\mathcal{F}$ with verifiers $\{v_f\}$ and parser ρ; held-out pairs H; held-out retries N; reinforcement pool $\mathcal{D}_{\mathrm{RL}}$ (no prompt contains a pair in H; structure share 33%); reward weight λ; group size G; β, $\varepsilon_{\mathrm{clip}}$, ϵ; steps T; learning-rate schedule η_s

Output: trained policy π_ϕ

$\pi_{\mathrm{ref}} \leftarrow \pi_{\phi_0}$

if *u is enabled* **then**

  `// Optional self-labelled supervised fine-tuning (§3.4)`

  $\mathcal{D} \leftarrow \emptyset$

  foreach $y \in \mathcal{Y}$ **do**

    $C(y) \leftarrow$ NULL; draw $k \sim \mathcal{U}\{1,2,3\}$

    for *attempt* $= 1, \ldots, N$ **do**

      sample $F \subseteq \mathcal{F}_y$ (families measurable on y), $|F| = \min(k, |\mathcal{F}_y|)$

      $C \leftarrow \{(f, \theta_f(y)) : f \in F\}$ with $\theta_f(y) = p_f(\rho(y))$ read off $\rho(y)$

      if *C contains no pair in H* **then**

        $C(y) \leftarrow C$; **break**

    end

    if $C(y) = \mathit{NULL}$ **then**

      $C(y) \leftarrow$ one constraint drawn from y's properties `// singleton: trivially held-out-free`

    end

    if $A(y, C(y)) = 1$ **then**

      $\mathcal{D} \leftarrow \mathcal{D} \cup \{(x_{C(y)}, y)\}$ `// else this draw of` y `is discarded`

    end

    `// with prob. 0.2: a structure-only pair from` y`'s first two 4-bar units, re-verified the same way`

  end

  $\pi_{\mathrm{ref}} \leftarrow$ fine-tune π_{ϕ_0} on $\mathcal{D}$

end

Algorithm 1: MusicRLVR (continued): verifier-driven GRPO

$\pi_\phi \leftarrow \pi_{\mathrm{ref}}$

`// Verifier-driven GRPO (§3.4)`

for *step* $s = 1, \ldots, T$ **do**

  $\phi_{\mathrm{old}} \leftarrow \phi$ (detached)

  sample a batch of prompts $x_C, C \sim \mathcal{D}_{\mathrm{RL}}$

  foreach x_C *in the batch* **do**

    sample a group $\{y_1, \ldots, y_G\} \sim \pi_{\phi_{\mathrm{old}}}(\cdot \mid x_C)$

    score every y_i by $R(y_i, C)$ (Equation 5)

    $\hat{A}_i \leftarrow \big(R(y_i, C) - \mu_C\big)/(\sigma_C + \epsilon)$ for every i (Equation 4) `// ` $\hat{A}_i = 0$ `for every` i `if` $\sigma_C = 0$

  end

  accumulate $\nabla_\phi(-\mathcal{J}(\phi))$ over the batch, clip its norm to 1.0, AdamW step at η_s `// Equation 6`

end

disabling the NZ tensor layout (it corrupts weight synchronisation during RL) and capping the rollout engine's context length, which otherwise reserves key-value cache for the model's full 262,144-token window.

B.2 Training Hyperparameters

In Table 7, we list the settings of the optional SFT, the matched GRPO runs of Table 2 and the seed-42 diagnostic GRPO run; both GRPO configurations keep 48 completions per update. The saved trainer configurations carry the remaining optimiser defaults.

Figure 6 shows the supervised stage, run for a fixed two-epoch budget. Two diagnostics from the GRPO logs are worth recording alongside it. The clipped fraction of Equation 6 is identically zero at all 300 steps, as expected when one optimiser update follows each rollout batch, so the clipping range never binds. In the default diagnostic GRPO run no completion reaches the 768-token cap: the longest sampled completion over training is 492 tokens, and the trainer's truncation counter stays at zero.

B.3 Second Base Model

In this section, we describe the Qwen3-8B run, which applies the Direct recipe unchanged: $\lambda = 0.7$, seed 43, 300 updates of 48 completions with $G = 8$, learning rate 10^{-6}, $\beta = 0.04$ and a 768-token completion cap. Qwen3-8B is a hybrid reasoning model; training and evaluation use its chat template with thinking disabled, as does the zero-shot Qwen3-8B row of Table 2. The larger model trains on two policy devices with ZeRO-3 and CPU offload of parameters and optimiser state (micro-batch 1, accumulation 24), which changes memory layout but not the update; a further device serves rollouts. The run takes 15.7 hours.

Where the two base models differ. The 4B base is the instruction-tuned Qwen3-4B-Instruct-2507 release, whereas Qwen3-8B is the original hybrid release used with thinking disabled; zero-shot, the smaller model is the stronger of the two (0.160 against 0.123 on Mixed). After training, the 4B and 8B policies reach 0.807 and 0.657 on Mixed. On the Structure suite the 8B policy reaches 0.900 (108/120), despite its base parsing only 19 of the 120 prompts, showing that the same Direct recipe can recover strong form control from a weak zero-shot starting point. The 4B policy remains stronger on Structure at 0.967, while the 8B policy posts the highest Unseen and Dense rates in Table 2, supporting transfer of the same recipe across model

Table 7: Hyperparameters of the optional SFT, the matched GRPO runs (seed 43) and the seed-42 diagnostic GRPO run. TP: tensor-parallel degree; mem. frac.: the rollout engine's memory fraction.

	Optional SFT	GRPO (matched)	GRPO (diagnostic)
Base model		Qwen3-4B-Instruct-2507	
Parameters trained	Full	Full	Full
Training data	12,890 pairs	6,000 prompts	6,000 prompts
Optimiser	AdamW	AdamW	AdamW
Learning rate	1×10^{-5}	1×10^{-6}	1×10^{-6}
LR schedule	Cosine	Cosine	Cosine
Warmup (fraction of steps)	3%	2%	2%
Batch size (per device $\times$ devices)	4×3 pairs	2×2 completions	2×3 completions
Gradient accumulation	4	12	8
Effective batch per optimiser step	48 pairs	48 completions	48 completions
Epochs / steps	2 epochs (538 steps)	300 steps	300 steps
Max sequence / completion length	1536 / –	1536 / 768	1536 / 768
Group size (G)	–	8	8
KL coefficient (β)	–	0.04	0.04
Clipping range ($\varepsilon_{\text{clip}}$)	–	0.2	0.2
Rollout temperature	–	1.0	1.0
λ (Equation 5)	–	0.7	0.7
Structure share in data	16.5%	33.1% (upsampled from 21%)	33.1%
Precision	bf16	bf16	bf16
Devices	3	2 train + 1 rollout	3 train + 1 rollout
Rollout engine (vLLM)	–	TP = 1, mem. frac. 0.92	TP = 1, mem. frac. 0.9
Framework	ms-swift 4.2.3	ms-swift 4.2.3	ms-swift 4.2.3
Wall time	∼45 min	∼3.7 hours	∼3 hours

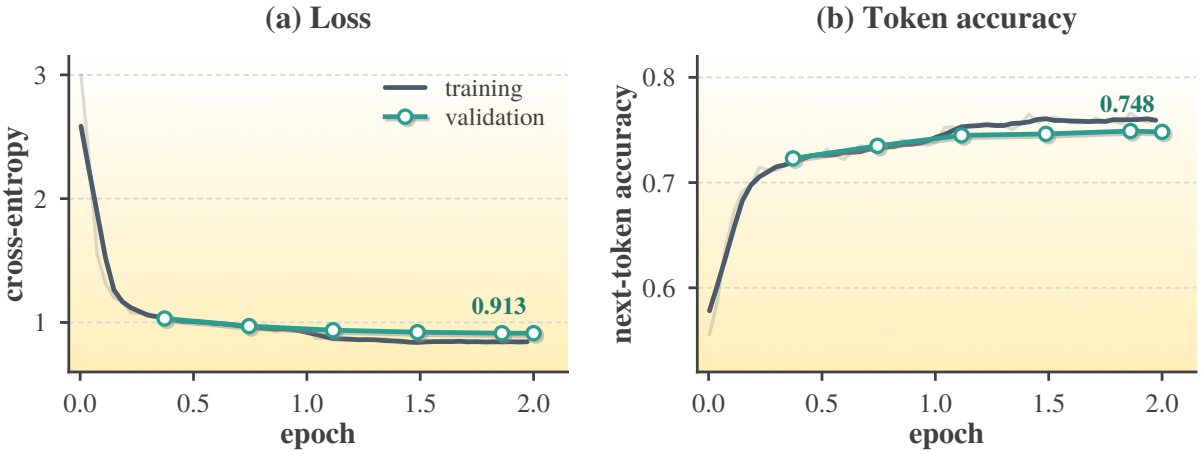


Figure 6: **The supervised stage.** Cross-entropy (a) and next-token accuracy (b) over the two SFT epochs, for the 12,890 training pairs and the 125-pair validation split. Validation loss decreases to 0.913 and validation accuracy plateaus at 0.748.

scales.

B.4 SFT Data Construction

In this section, we describe how the supervised training data is built from the IrishMAN corpus (Wu et al., 2023), a collection of ∼45,000 traditional Irish tunes in ABC notation. The construction pipeline has four steps:

1. **Filter and parse.** Tunes whose body contains the builder's list of unsupported tokens, or that have fewer than 4 or more than 32 written bars, are discarded; the survivors are parsed with `music21` and kept only if the parsed bar count agrees with the written one.
2. **Property extraction.** The same verifiers used for evaluation extract the true properties of each tune: its key (hybrid policy), meter, bar count, pitch range, final note, the set of distinct note durations, and the largest melodic interval.
3. **Instruction rendering.** A random subset of the extracted properties is selected (with k sampled uniformly from $\{1, 2, 3\}$, capped by the number of measurable properties) and rendered into a constraint instruction using the same template as evaluation. The tune itself is the target output.
4. **Verification gate.** Every (instruction, tune) pair is re-verified: the tune is checked against the rendered constraints using the evaluation verifiers. Any pair that fails verification is

discarded. This ensures that no training example teaches the model to produce output that violates its own constraints.

Form examples (`structure` family) are constructed separately: for a tune with at least eight bars whose first two four-bar units differ, the two units are arranged into a sampled AABA/AABB/ABAB shape (attempted with probability 0.2 per tune; the resulting pair carries the structure constraint only). The resulting 12,890 pairs form the training set, with a further 125 pairs held out for validation. Importantly, the four held-out family pairs (§4) are enforced during construction: any pair whose constraints contain a held-out combination is regenerated with a different family subset.

B.5 SFT Training Examples

In this section, we present three representative (instruction, output) pairs from the SFT training set to illustrate the data quality and the range of constraint combinations.

B.6 Prompt Template

In this section, we present the fixed template that renders every benchmark item into a prompt. The shared instruction prefix (the first paragraph) is part of the user message of every item; the constraint block is populated from the item's constraint list. No chain-of-thought, few-shot examples, or reasoning instructions are given.

```
Write a single monophonic tune in ABC
notation.
Rules: write every bar out
explicitly (no repeat signs like |:
or :|), one voice only, no chords, no
lyrics, no grace notes. Start with
an X: header, include T:, M:, L:, K:
headers, then the music. Output ONLY
the ABC notation and nothing else.

The tune must satisfy ALL of the
following constraints:
- ⟨constraint 1 in natural language⟩
- ⟨constraint 2 in natural language⟩
  ⋮
```

The constraint text for each family is generated by a deterministic renderer that maps parameters to a fixed English sentence. For example, the `key` family with $\theta = (\mathrm{D}, \mathrm{major})$ renders as "*the tune must be in D major*"; the `structure` family with $\theta = (\mathrm{AABA}, 4)$ renders as "*the tune must be exactly 16 bars with an AABA form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ*". The paraphrase suite (§4) replaces these renderers with alternative phrasings while preserving the constraint values.

B.7 Dataset Statistics

In this section, we report the composition of the benchmark splits and the training data.

Benchmark splits. Table 8 summarises the composition of the six original benchmark splits. The mixed suite spans $k \in \{1, 2, 3, 4\}$ and covers all eight families; the structure split is the only one where every item includes the cross-section form `structure` family. Family frequencies depend on the sampler and exclusion rules (Table 12); the structure split contains its namesake family in every item and one of five companion families in 66 of 120.

Training data. The supervised stage uses 12,890 instruction–tune pairs self-labelled from the IrishMAN corpus. The reinforcement stage uses 6,000 synthetic prompts with $k \in \{1, 2, 3\}$ (2,046/3,060/894 respectively). After the reward-design iteration of Appendix D.15, the share of prompts containing the `structure` family was raised to 33.1% (1,983/6,000) to increase exposure to form constraints; every other family appears in 1,068–1,347 prompts.

B.8 Prompt Coverage, Training Overlap, and Split Composition

In this section, we analyse prompt coverage, overlap with the training prompts and split composition. The weighting and fixed-k comparisons below use the diagnostic SFT-initialised reference; the suite definitions and prompt counts apply to both studies.

The 2,180 suite items are sampled and contain 1,623 distinct prompt strings across all eight splits. Table 9 compares each prompt with the user messages in both training datasets. "Set overlap" ignores constraint order and compares family–parameter sets instead. Unseen, High-k and Dense share no complete prompt or constraint set with either training stage, and neither stage requests any of the four held-out pairs. Single and Structure prompts all occur in training.

Equal weighting of distinct prompts. Table 10 first averages repeated rows for each prompt within one evaluation, then averages over distinct prompts. The RL column uses the same fixed checkpoint and evaluation files as Table 44. The advantage over

SFT example 1: single constraint (`meter`).

Instruction: Write a single monophonic tune in ABC notation . . . The tune must satisfy ALL of the following constraints: – the meter must be 6/8, and every bar must contain exactly the right total duration.

Target (from IrishMAN):

```
X:89124
T:Tune
L:1/8
M:6/8
K:G
GBd gdB | GBd cBA | GBd gdB | cBA edc | GBd gdB | GBd cBA | cBA dcB | cAF G3 |
DFA cBA | GAB d2 d | DFA cBA | BAG AGE | DFA cBA | GAB d2 d | cBc dcB | cAF G3 |
```

Every one of the 16 bars contains exactly six eighth-note units, and the tune carries no repeat sign, so it passes the gate and the meter verifier.

SFT example 2: two constraints (`length` + `key`).

Instruction: . . . – the tune must be exactly 32 bars long (no pickup bar). – the tune must be in D major.

Target:

```
X:106003
T:Tune
L:1/8
Q:1/8=180
M:6/8
K:D
DFA cdc | AFA BGE | DFA dcd | AFA gfe | DFA cdc | AFA BGE | afd bge | AFD E2 D |
DFA cdc | AFA BGE | DFA dcd | AFA gfe | DFA cdc | AFA BGE | afd bge | AFD E2 D |
fef gfg | afd AFD | fef gfg | DFA B2 A | fdf gfg | afd bge | fga Bcd | AFD E2 D |
fef gfg | afd AFD | fef gfg | DFA B2 A | fdf gfg | afd bge | fga Bcd | AFD E2 D |
```

The header reads `K:D` and content analysis returns D major; exactly 32 bars. The constraint parameters were read off this tune's own properties, and the pair was re-verified before admission.

SFT example 3: structure constraint (`structure`, AABA).

Instruction: . . . – the tune must be exactly 16 bars with an AABA form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ.

Target (assembled from the source tune's first two 4-bar units):

```
X:141183
T:Tune
L:1/4
M:3/4
K:Bmin
B2 c | d3 | d2 a | f e d | B2 c | d3 | d2 a | f e d |
c3 | e3 | f2 B | B3 | B2 c | d3 | d2 a | f e d |
```

Units 1, 2, and 4 are identical (A sections, bars 1–4, 5–8, 13–16); unit 3 (bars 9–12) differs, satisfying the AABA form. The two units are the source tune's first eight bars; every assembled pair is verified before entering the dataset.

Table 8: Composition of MusicConstraintBench splits: number of items, distribution over k (number of simultaneous constraints), and family coverage, as ranges where shown.

Split	n	k distribution	Families (count)
Mixed	300	1:100, 2:100, 3:60, 4:40	All eight
Single	200	1:200	All eight (16–29 each)
Seen comb.	200	2:116, 3:84	All eight (51–72 each)
Unseen comb.	200	2:88, 3:112	All eight (54–70 each)
High-k	160	4:160	All eight (56–104 each)
Structure	120	1:54, 2:66	`structure`: 120; 5 others

SFT persists on all eight splits under this weighting. On Structure, whose 120 items repeat 50 prompts, the rates are 0.637 and 0.522, respectively. Item-paired tests in the main analysis retain the original sampler weights, so they measure the sampled suites rather than 2,180 distinct instructions. Table 11 applies the same weighting to the three matched checkpoints of Table 2, read from the same saved per-item files.

Fixed k does not fix family composition. Table 12 gives family frequencies in High-k, Dense and Edge. The first two share $k = 4$ but are

Table 9: Coverage and overlap with either training stage. Overlap columns count sampled test rows, while "unique prompts" counts distinct strings.

Split	Items	Unique prompts	Exact overlap	Set overlap
Mixed	300	222	201	203
Single	200	39	200	200
Seen	200	177	147	151
Unseen	200	159	0	0
High-k	160	156	0	0
Structure	120	50	120	120
Dense	500	490	0	0
Edge	500	437	32	39

Table 10: Original item weighting and equal weighting of unique prompts, using the main-table criterion and the diagnostic reference checkpoint.

Split	SFT, items	SFT, unique	RL, items	RL, unique
Mixed	0.477	0.402	0.663	0.615
Single	0.745	0.705	0.835	0.820
Seen	0.415	0.407	0.635	0.618
Unseen	0.440	0.396	0.595	0.547
High-k	0.231	0.237	0.356	0.359
Structure	0.600	0.522	0.808	0.637
Dense	0.290	0.290	0.370	0.371
Edge	0.076	0.078	0.162	0.164

independently sampled and have different family mixtures. Edge additionally mixes pair-free and held-out items ($171/500$ contain a held-out pair). Appendix D.11 scores the pair-free and held-out-pair halves of Edge separately. Table 13 bootstraps the Dense-minus-High-k difference for each model, separating item-sampling variation from this composition difference.

All three intervals include zero: at $k = 4$, requiring a held-out pair does not measurably lower compliance for any of the three models. Table 12 gives the family composition the two splits differ in.

B.9 Reproducibility

In this section, we list the scripts that regenerate the paper's tables, macros and figures. `make_assets.py` regenerates the original quantitative tables and numerical macros; `make_audit_assets.py` adds the overlap, sensitivity and gate-audit tables; `src/rescore_subsets.py` produces the Edge-subset and stricter-key tables. The historical reporting pass, `make_checkpoint_assets.py -sync`, computes reference-table, by-k, Edge-subset and fixed-k comparisons from saved verdicts, checks ablation equivalence under the format audit, and supplies the fixed-checkpoint decompositions; it also supplies the corresponding historical figure panels. The matched tables and macros are generated by `make_matched_s43_assets.py` after all three runs finish, and the pooled by-k figure by `make_fig_degrade_pooled.py`. This script checks saved per-item IDs, success counts and provenance hashes for all 24 evaluations, then renders the two paired comparisons with eight-suite Holm correction. Figure scripts read the same saved reports or training logs; each later analysis (reranking, k ladder, Edge factors, output variety) names its script where it is reported. The family-definition and rendering tables are maintained descriptions of the sampler. The six original suites were used during reward development; Dense and Edge were added afterwards, without that exposure. All training and local evaluation ran on Ascend 910B2 accelerators (64 GB HBM each, CANN 8.2.RC1) with vLLM 0.11.0 through `vllm-ascend`; the seed-42 diagnostic GRPO runs used three policy accelerators and one rollout accelerator. The matched study used two policy accelerators and one rollout accelerator. Local models were served at context 4,096 and generated at most 1,200 tokens per item, greedy. The hosted baselines were queried at temperature 0 through the NVIDIA NIM API (`integrate.api.nvidia.com`), MiniMax-M3 as `minimaxai/minimax-m3` in August 2026, with per-model completion budgets of 800 tokens for Llama-3.1-70B, 4,000 for MiniMax-M3 and 2,048 of context for ChatMusician-7B; budgets and timeouts per model are in the release scripts. The release includes per-item completions, original and strict gate decisions, aggregate reports, audit scripts, and training logs. Historical reports remain unchanged; `src/verifiers_legacy.py` reproduces the training and main evaluation metric, while `src/verifiers.py` implements the stricter gate. The reward plugin pins the historical gate unless strict-gate training is explicitly selected. `src/run_verify.py` defaults to this same A criterion (`legacy` mode).

B.10 Evaluation Nondeterminism

In this section, we quantify evaluation nondeterminism. Compliance evaluation uses temperature 0, yet vLLM's batched decoding is not bitwise reproducible: evaluating the diagnostic checkpoint (`outputs/grpo_4b_v2/checkpoint-300`) twice, on 08-22 and 08-24, with identical serving flags (Table 15), none of the 160 high-k completions was byte-identical across the two runs. A plausible mechanism is that the ordering of floating-point reductions depends on batch composition, and that the first generated token (the arbitrary `X:` reference number) sits on a near-tie

Table 11: The same weighting applied to the three matched checkpoints of Table 2: "items" repeats the sampler weights of Table 2, "unique" averages within each distinct prompt first. The splits whose prompts repeat most, Single (39 distinct prompts) and Structure (50), move most; the ordering between the three checkpoints is unchanged on every split but Edge, where the two Direct arms swap by 0.008.

	Direct GRPO ($R = A$)		MusicRLVR (SFT init.)		MusicRLVR (Direct)	
Split	items	unique	items	unique	items	unique
Mixed	0.797	0.741	0.577	0.527	0.807	0.760
Single	0.960	0.949	0.880	0.875	0.985	0.981
Seen	0.745	0.729	0.585	0.567	0.815	0.814
Unseen	0.565	0.509	0.530	0.479	0.595	0.547
High-k	0.469	0.458	0.350	0.353	0.575	0.571
Structure	0.908	0.860	0.383	0.457	0.967	0.940
Dense	0.322	0.322	0.366	0.363	0.362	0.358
Edge	0.300	0.291	0.134	0.135	0.310	0.283

Table 12: Number and percentage of items requesting each family. Percentages sum to more than 100% because items contain multiple families.

Family	High-k ($n = 160$)	Dense ($n = 500$)	Edge ($n = 500$)
`key`	76 (47.5%)	288 (57.6%)	168 (33.6%)
`meter`	82 (51.2%)	282 (56.4%)	160 (32.0%)
`length`	60 (37.5%)	209 (41.8%)	127 (25.4%)
`range`	100 (62.5%)	236 (47.2%)	164 (32.8%)
`final`	104 (65.0%)	234 (46.8%)	180 (36.0%)
`rhythm`	84 (52.5%)	279 (55.8%)	152 (30.4%)
`leap`	78 (48.8%)	270 (54.0%)	157 (31.4%)
`structure`	56 (35.0%)	202 (40.4%)	148 (29.6%)

between many candidates, so a small numerical difference there can send the whole continuation down a different path.

The two evaluations differ by 0.040 on Mixed and 0.050 on Structure.

The first evaluation supplies the historical reference and diagnostic values except Dense and Edge, which come from the re-evaluation, and Figure 7(d), which uses a separate full-retention run. The fixed-k comparison uses the re-evaluated High-k score ($0.394 \rightarrow 0.370$).

B.11 Data and Code Release

We will release the code and synthetic benchmark suites under an MIT licence, with corpus-derived data and model artefacts subject to their upstream terms:

- **MusicConstraintBench benchmark suites:** eight frozen JSONL files containing all 2,180 benchmark items plus the 400-item paraphrase suite. Each item includes the prompt and the constraint specifications; the generation seed is a parameter of the suite sampler rather than a per-row field.
- **Verifier code:** the eight constraint verifiers, the ABC parser, the reward function, the historical verifier, the corrected strict gate, and both sets of scoring reports.
- **SFT and RL data:** the 12,890 supervised training pairs and 6,000 RL prompts.
- **Model outputs:** per-item completions and per-item verifier decisions (with all sub-signals) for every checkpoint and baseline evaluated in this paper.
- **Evaluation reports:** aggregate reports, significance tests, quality statistics, prompt-overlap and unique-prompt analyses, strict-gate audits, and the training logs from which the reward curves are drawn.
- **Asset generation scripts:** `make_assets.py` for the initial tables, `make_audit_assets.py` for supplementary analyses, the final reporting pass `make_checkpoint_assets.py -sync`, and one `make_fig_*.py` per figure, all reading saved reports, so every number in the paper is traceable to a specific evaluation output.

The IrishMAN corpus (Wu et al., 2023) used for SFT training is publicly available; source-specific rights and attribution should be preserved for corpus-derived targets. Model distribution follows the licence of the exact Qwen3-4B-Instruct-2507 checkpoint (Yang et al., 2025). No human subjects were involved in this research; all evaluation is automated.

Table 13: Dense minus High-k: absolute all-satisfied rate differences with percentile intervals from 20,000 within-split item bootstrap replicates (seed 20260909).

Model	High-k	Dense	Δ	95% interval
SFT	0.231	0.290	$+0.059$	$[-0.019, +0.132]$
MusicRLVR (re-evaluated)	0.394	0.370	-0.024	$[-0.111, +0.062]$
Llama-3.1-70B	0.056	0.040	-0.016	$[-0.057, +0.021]$

Table 14: Parameter values of the Edge split against the standard suites. Families are sampled by the same generator; only the value sets differ. `key`, `final` and `structure` take the same values in both, so they are omitted.

Family	Standard suites	Edge
`meter`	2/4, 3/4, 4/4, 6/8	5/4, 2/2, 9/8, 12/8
`length`	8, 12, 16 bars	4, 24, 32 bars
`leap`	$\leq$ 7, 9, 12 semitones	$\leq$ 2, 3, 4 semitones
`rhythm`	three- and two-duration sets	{16th, 8th}, {quarter, half}, {quarter}, {half}
`range`	mixed spans	one-octave spans (C4–C5, D4–D5, E4–E5, G3–G4, A3–A4)

Table 15: The same checkpoint evaluated twice under greedy decoding. "Flipped" counts items whose binary all-satisfied outcome differed between the two. Both have parse rate 1.000.

Split	n	First	Second	Flipped	0→1	1→0
Mixed	300	0.663	0.703	52 (17.3%)	32	20
High-k	160	0.356	0.394	32 (20.0%)	19	13
Structure	120	0.808	0.758	26 (21.7%)	10	16

C Benchmark and Verifier Details

In this section, we present the verifier implementation and its audit, the key calibration, the held-out pairs, ABC notation and examples of the prompt renderings.

C.1 Verifier Implementation Details

In this section, we describe the validation gate and the family verifiers. The historical gate ρ, which is the validation gate of §3 and is also called the parse gate below, is used in training and in all tables labelled A, and runs first. It extracts the ABC block from the completion and rejects the output if extraction fails, if the text contains `|:` or `:|`, if `music21` raises or exceeds a 20-second parse timeout, or if the selected first part has fewer than eight notes, fewer than four non-empty measures, or any chord object. Outputs that clear the gate are then scored by the family verifiers. Appendix C.2 adds stricter format checks on the same saved completions, under which the matched checkpoints of Table 2 keep every success.

Each of the eight constraint families is implemented as a deterministic function that takes a `music21 Score` object and the constraint parameters, returning a binary verdict. We describe the non-trivial implementation choices below.

`key`. As discussed in §3.2, the verifier uses a hybrid policy: the `K:` header must match the request, *and* key-profile analysis of the note content (`score.analyze("key")`, which in `music21` 10.5.0 selects the Aarden–Essen profiles) must return the requested key or its relative major/minor. On a calibration set of 400 IrishMAN tunes, this hybrid policy achieves 0.890 agreement with the corpus key annotation, compared to 0.750 for content analysis alone and 1.000 (trivially) for header-only checking.

`meter`. The verifier checks two conditions: (1) the time signature in the `M:` header matches the request, and (2) every non-empty bar's total duration (notes and rests, in quarter-note equivalents) equals that signature's bar length within 10^{-4}. Incomplete initial or final bars receive no tolerance.

`length`. The number of non-empty bars is counted after parsing and compared with the request. The prompt's "no pickup bar" is enforced here as well: the first and last written bar must each carry a full measure of the tune's own time signature, so an output that opens with an anacrusis and closes with its complement fails even when no `meter` constraint is requested.

`range`. Every sounding pitch is compared against the specified minimum and maximum (as MIDI numbers). Rests are ignored.

`final`. The pitch class of the last sounding note is compared against the request; octave is always ignored.

`rhythm`. Every note's duration in quarter-note units, rounded to four decimals, must belong to

the admissible set, and any rest fails the check. A dotted or tied value therefore passes only if its parsed duration is itself admissible.

`leap`. Every pair of consecutive sounding pitches is examined; the verifier returns 0 if any interval exceeds the stated maximum (in semitones).

`structure`. A tune with other than sixteen non-empty bars fails outright; otherwise it is split into four units of four bars each. Each bar is reduced to its signature (event type, MIDI pitch and duration rounded to four decimals, rests included). The minimal pair sets are AABA: $E = \{(1,2),(1,4)\}$, $D = \{(1,3)\}$; AABB: $E = \{(1,2),(3,4)\}$, $D = \{(1,3)\}$; ABAB: $E = \{(1,3),(2,4)\}$, $D = \{(1,2)\}$. For each pair in E (Equation 2) the verifier compares the bar signatures position by position; for each pair in D it checks that at least one position differs. The graded training signal $\bar{e}$ is the fraction of bar positions that match across all pairs in E; the binary evaluation signal requires all to match.

Validation without annotators. Except for the hybrid `key` check, every family is a deterministic function of the parsed score, so the question is whether the implementation computes the property it claims. `src/validate_verifiers.py` builds 32 ABC scores whose property values are fixed by construction (`key` 2, `meter` 4, `length` 3, `range` 5, `final` 3, `rhythm` 5, `leap` 5, `structure` 5, including the inclusive bounds) and all 32 verdicts are correct. It then starts from one tune that satisfies six families at once and mutates one property at a time (`length`, `meter`, `range`, `final`, `rhythm`, `leap`); each mutation flips exactly the targeted verdict and leaves the other five unchanged. Three conventions are reported rather than asserted: a pickup bar counts against `length`, an enharmonic spelling of the final pitch class is accepted, and a tune whose notes outline the relative key of its header is accepted by `key`. `src/validate_meter_corpus.py` runs the meter check on 400 IrishMAN tunes: the parsed time signature equals the declared `M:` field on all 400.

C.2 Stricter Fixed-Instruction Gate Audit

In this section, we audit the diagnostic checkpoints of Table 44 and the three matched RL checkpoints of Table 2 under stricter format checks.

The historical gate can accept a second voice because most checks inspect only the first part, whereas key analysis inspects the whole score. It also does not reject every wrapper, missing header, lyric or grace note. We retain that verifier as `verifiers_legacy.py` to reproduce the saved training runs, and implement a stricter gate in `verifiers.py`. It requires a single tune with the specified headers, no unconsumed wrapper text, no repeat notation, lyrics or grace notes, and no multiple parts, multiple voices, chord objects or overlapping notes. Whitespace is normalised when detecting wrappers. The eight family checks remain unchanged. This hardens the fixed-instruction checks.

The stricter metric is

$$A_{\text{strict}}(y, C) = A(y, C)\, \mathbb{I}[g_{\text{strict}}(y)].$$

`audit_strict_gate.py` evaluates the additional gate on saved completions; no regeneration or retraining is involved. For the configurations within this manuscript's scope, the audit covers 175 evaluation files and 43920 rows: the three matched checkpoints of Table 2 on all eight suites, plus the baseline, training-configuration, ablation, sweep, paraphrase and retained dynamics checkpoints. All file-level outcomes, rejection reasons and source hashes are saved separately. Tables 16–17 show the original six suites; Table 18 adds Dense and Edge. Across the audit subset, 201 previously successful rows fail the stricter gate, and *none of them belongs to the three matched RL rows of Table 2*: those three checkpoints incur no additional gate rejection and lose no success on any of the 6,540 items they were scored on, so every rate in those three RL rows is unchanged under A_{strict}. The main comparison and reward curves use A; this audit measures the additional fixed-format requirements, which are not included in the main score.

The SFT and fixed SFT-to-GRPO checkpoints incur no additional gate rejections on any of the eight suites. ChatMusician's strict success rates are zero because its otherwise accepted outputs omit required headers or include wrappers. This audit therefore changes some baseline scores; a strict-format failure is a formatting verdict.

The corrected gate has regression checks for valid monophonic output and for multiple voices, chords, grace notes, lyrics, missing title, prose wrappers and repeat signs. Historical training uses the frozen gate.

Table 16: All-satisfied rate under the stricter fixed-instruction gate A_{strict} on the original-suite completions. The last three rows are the matched checkpoints of Table 2; every one of their rates equals the corresponding A rate.

Model	Mixed	Single	Seen	Unseen	High-k	Structure
Qwen3-4B	0.160	0.225	0.100	0.040	0.037	0.100
Qwen3-8B	0.120	0.210	0.115	0.050	0.006	0.000
Qwen3-32B	0.357	0.535	0.240	0.190	0.094	0.267
Llama-3.1-70B	0.377	0.695	0.300	0.190	0.050	0.167
MiniMax-M3	0.367	0.580	0.315	0.190	0.100	0.008
ChatMusician-7B	0.000	0.000	0.000	0.000	0.000	0.000
SFT	0.477	0.745	0.415	0.440	0.231	0.600
MusicRLVR (diagnostic reference)	0.663	0.835	0.635	0.595	0.356	0.808
Direct GRPO ($R = A$)	0.797	0.960	0.745	0.565	0.469	0.908
MusicRLVR (Direct)	0.807	0.985	0.815	0.595	0.575	0.967
MusicRLVR (SFT init.)	0.577	0.880	0.585	0.530	0.350	0.383

Table 17: Additional gate rejections across the six original suites (1,180 rows per model). Lost successes previously had $A = 1$ and now have $A_{\text{strict}} = 0$; an extra rejection need not change success if a family already failed. Reasons list only the first detected violation per output.

Model	New rejects	Lost successes	Leading reasons
Qwen3-4B	7	0	non abc wrapper or unconsumed text: 7
Qwen3-8B	6	1	non abc wrapper or unconsumed text: 5, grace notes forbidden: 1
Qwen3-32B	22	4	non abc wrapper or unconsumed text: 19, repeat signs forbidden: 2
Llama-3.1-70B	26	3	non abc wrapper or unconsumed text: 26
MiniMax-M3	1	0	non abc wrapper or unconsumed text: 1
ChatMusician-7B	749	151	missing required header: 734, non abc wrapper or unconsumed text: 15
SFT	0	0	–
MusicRLVR (diagnostic reference)	0	0	–
Direct GRPO ($R = A$)	0	0	–
MusicRLVR (Direct)	0	0	–
MusicRLVR (SFT init.)	0	0	–

Table 18: Historical and strict all-satisfied rates on the two additional suites, using the same saved completions.

Model	Dense A	Dense A_{strict}	Edge A	Edge A_{strict}
Qwen3-4B	0.008	0.008	0.010	0.010
Qwen3-8B	0.020	0.020	0.032	0.032
Qwen3-32B	0.074	0.074	0.112	0.102
Llama-3.1-70B	0.040	0.038	0.058	0.056
MiniMax-M3	0.088	0.088	0.160	0.160
ChatMusician-7B	0.020	0.000	0.046	0.000
SFT	0.290	0.290	0.076	0.076
MusicRLVR (diagnostic, re-evaluated)	0.370	0.370	0.162	0.162
Direct GRPO ($R = A$)	0.322	0.322	0.300	0.300
MusicRLVR (Direct)	0.362	0.362	0.310	0.310
MusicRLVR (SFT init.)	0.366	0.366	0.134	0.134

C.3 Key Verifier Calibration

In this section, we provide the full calibration data of the hybrid key verifier (§3.2), the most complex verifier in our suite, which involves both header matching and content-based key analysis.

Calibration dataset. We sampled 400 tunes from the IrishMAN corpus at random (seeded), each with a key annotation in the `K:` header. We then ran four key-verification strategies and measured their agreement with that annotation (Table 19):
The reference here is the corpus `K:` annotation, which the header-only policy reproduces by construction. The hybrid policy matches the content-with-tolerance policy on this set; its advantage appears when the model might game the header, since a policy that prints `K:G` above music in C major passes the header-only check but fails the hybrid one. The remaining 0.110 is disagreement between the analysis routine and that annotation, which modal tunes, extensive chromaticism and very short tunes can each move. The same policy applies to every model we score.

Table 19: Key verifier calibration on 400 IrishMAN tunes. Agreement is the fraction of tunes where the verifier's answer matches the corpus key annotation.

Strategy	Agreement
Header only (trivially gameable)	1.000
Content analysis only (Aarden–Essen profiles)	0.750
Content analysis with relative-key tolerance	0.890
Hybrid: header AND content $\in$ {key, relative}	0.890

Effect on reported numbers. The key family's per-constraint satisfaction rate (Table 50) is 0.862

on the single split and 0.671 on the high-k split. If we used a header-only verifier, these would be higher (the model rarely writes the wrong `K:` header), but header-only acceptance does not inspect the notes. Content analysis with relative-key tolerance carries the 0.110 disagreement above; exact content-only matching disagrees on 0.250 of the calibration tunes. We adopt the hybrid policy and log all sub-signals, so future work can re-score under any policy.

C.4 Held-out Pair Rationale

In this section, we explain how the held-out pairs are chosen. The four held-out pairs (`key`, `rhythm`), (`meter`, `leap`), (`final`, `structure`) and (`length`, `range`) were chosen to form a *perfect matching* on the eight families, ensuring that (1) every family participates in exactly one held-out pair, (2) no family appears in multiple exclusions, and (3) the matching constrains the maximum testable k to 4 without touching any held-out pair, which naturally defines the high-k split.

The specific pairing was selected to maximise the diversity of inter-family interaction types:

- `key`–`rhythm`: one harmonic, one temporal, with no obvious interaction (a tune can be in any key with any rhythm vocabulary).
- `meter`–`leap`: one temporal, one melodic; the meter determines bar duration but should not affect interval size.
- `final`–`structure`: one local (last note), one global (16-bar form); the most interesting test of whether a model can coordinate endpoint and large-scale structure without having seen them together.
- `length`–`range`: both are "bounding" constraints (number of bars, pitch boundaries) that interact primarily through the amount of melodic material available.

The supervised dataset contains zero instances of any held-out pair, verified by an exhaustive check over all $12{,}890$ pairs before training.

C.5 ABC Notation Primer

In this section, we introduce ABC notation, a plain-text format for monophonic music widely used in folk-music archives. A minimal valid tune contains a header block and a body:

```
X:1                % reference number
T:My Tune           % title
M:4/4               % meter (time signature)
L:1/8               % default note length
K:G                 % key signature
G2 BG dGBG | G2 Bd cAFA | G2 BG dGBG | cABG AGFG |
```

Key conventions: note names `C D E F G A B` denote the octave starting at middle C; lowercase `c d e f g a b` denote one octave higher; trailing apostrophes (e.g. `c'`) raise further. A digit after a note name multiplies its duration by that factor relative to the default length `L:`. Thus `G2` in a tune with `L:1/8` is a quarter note on G. The pipe character `|` separates bars.

Because ABC is plain text, a language model generates it token by token in exactly the way it generates code or prose. The `music21` library (Cuthbert and Ariza, 2010) parses ABC into a rich internal representation (pitch, duration, key analysis, interval computation) that our verifiers operate on. If parsing fails, for instance because the model outputs prose instead of notation or uses an invalid header like `L:8` instead of `L:1/8`, the gate returns $\rho(y) = \bot$, so $A(y, C) = 0$ (Equation 1).

C.6 ABC Notation: A Worked Example

In this section, we provide a token-by-token walk-through of a model-generated output that passes all four constraints of a high-k problem, for readers unfamiliar with ABC notation.

Problem. Meter 3/4; exactly 8 bars; last note E; eighth and quarter notes only.

Model output:

```
X:17011
T:Tune
L:1/8
M:3/4
K:Edor
B2 B2 AG | E2 E2 DE | F2 F2 ED | B2 B2 AG |
E2 E2 DE | F2 F2 ED | B2 B2 AG | E2 E2 DE |
```

Token-by-token analysis:

- `X:17011`: Reference number (arbitrary, used by databases).
- `T:Tune`: Title (arbitrary, not constrained).
- `L:1/8`: Default note length is an eighth note. This means `B` alone is an eighth note, `B2` is two eighth notes = one quarter note.
- `M:3/4`: Time signature: three quarter notes per bar. Since `L:1/8`, each bar must contain six eighth-note equivalents. ✓ meter.
- `K:Edor`: Key of E Dorian (E minor with raised sixth). The key constraint is not active in this problem, so any key is acceptable.
- `B2 B2 AG`: Bar 1. `B2` = quarter note B (two eighths); `B2` = quarter note B; `A` = eighth note A; `G` = eighth note G. Total: $2 + 2 + 1 + 1 = 6$ eighth-note equivalents. ✓ meter. Durations used: quarter, eighth. ✓ rhythm.

- `|`: Bar line separator.
- `E2 E2 DE`: Bar 2. Same structure. Total 6. ✓
- . . . (bars 3–7 follow the same pattern) . . .
- `E2 E2 DE |`: Bar 8 (last bar). The final token is `E`, an eighth note on pitch E. ✓ final.

Verification summary: meter ✓ (all 8 bars have 6 eighth-note equivalents); length ✓ (exactly 8 bars); final ✓ (last note is E); rhythm ✓ (only eighth and quarter notes appear). All four constraints satisfied; $A = 1$.

C.7 Constraint Rendering Examples

In Table 21, we show the complete set of constraint-text renderers: for each family, the table lists the parameter types and one representative rendering. These renderers are deterministic functions, so every benchmark item with the same family and parameters produces the same constraint text.

The constraint texts are designed to be unambiguous to a musically literate reader. The `meter` renderer explicitly states "every bar must contain exactly the right total duration" because simply saying "the meter is 3/4" could be interpreted as a header-only requirement. Similarly, the `structure` renderer spells out the matching/differing requirement rather than assuming the model knows what AABA form means.

C.8 Paraphrase Examples

In Table 22, we show three examples of the paraphrase suite, which re-renders the same constraint values with alternative wordings; they illustrate the range of surface variation, and the constraint parameters are unchanged.

Table 20: Table 1 with the prompt column restored. Under each family name is the span it is checked over; the second column lists the default sampler's values (the verifier domain Θ_f is wider), the third the sentence the prompt template emits for one such value, verbatim from the renderer. `structure` is set apart as the only cross-section form family.

Family	Default sampler values	Prompt line (one sampled value)
key global	13 keys: 8 major, 5 minor	*the tune must be in D minor*
meter every bar	$4/4$, $3/4$, $6/8$, $2/4$; each bar must sum to the signature	*the meter must be 6/8, and every bar must contain exactly the right total duration*
length global	8, 12 or 16 bars, no pickup	*the tune must be exactly 16 bars long (no pickup bar)*
range every note	4 inclusive pitch spans, e.g. G3–A5	*every note must lie between G3 and A5 inclusive*
final last note	a pitch class, any octave (a scale degree 1/3/5 when a key is present)	*the last note of the tune must be E (any octave)*
rhythm every note	4 admissible duration sets; no dotted values, no rests	*use only eighth and quarter notes (no dotted notes, no other durations, no rests)*
leap adjacent notes	largest melodic interval: 7, 9 or 12 semitones	*melodic motion must be smooth: no melodic leap larger than a major sixth*
structure 4-bar units	AABA, AABB or ABAB over 16 bars; equal letters identical, different letters distinct	*the tune must be exactly 16 bars with an ABAB form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ*

Table 21: Constraint rendering for all eight families. The "Parameters" column shows the parameter space; the "Rendered text" column shows the template output for one example value. Paraphrase variants (Appendix C.8) replace these templates with alternative wordings.

Family	Parameters	Rendered text (example)
`key`	key name × mode	"The tune must be in G major."
`meter`	time signature	"The meter must be 3/4, and every bar must contain exactly the right total duration."
`length`	bar count	"The tune must be exactly 16 bars long (no pickup bar)."
`range`	low pitch, high pitch	"Every note must lie between C4 and C6 inclusive."
`final`	pitch class; any octave	"The last note of the tune must be D (any octave)."
`rhythm`	admissible duration set	"Use only eighth and quarter notes (no dotted notes, no other durations, no rests)."
`leap`	max interval name	"Melodic motion must be smooth: no melodic leap larger than a perfect fifth."
`structure`	form label, unit size	"The tune must be exactly 16 bars with an AABA form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ."

Table 22: Paraphrase examples. Each row shows the original constraint text and one paraphrased variant; the constraint parameters are identical.

Original	Paraphrased
The tune must be in A minor.	Compose the melody in the key of A minor.
The meter must be 4/4, and every bar must contain exactly the right total duration.	Write it in 4/4 time, and make every measure add up to exactly the right duration.
Melodic motion must be smooth: no melodic leap larger than an octave.	Keep the melodic motion conjunct: no interval between consecutive notes may exceed 12 semitones.

D Additional Experiments

In this section, we present additional experiments. Appendix D.13 contains the matched comparison; all other RL analyses in this section preserve the diagnostic study, with SFT initialisation unless a Direct row or another configuration is explicitly named, and Table 44 collects those historical reference scores.

Unless explicitly labelled otherwise, all-satisfied rates use A (Equation 1), as in the main comparison; per-family statistics report the individual verifier verdicts. Appendix C.2 separately reports the additional format-gate audit.

D.1 Per-Family Satisfaction Across Models

In Table 23, we break the all-satisfied rate of every model of Table 2 down by constraint family over all 2,180 items. Each cell is the fraction of items requesting the family whose output satisfies it; an output rejected by the validation gate fails every

Table 23: Per-family satisfaction over all 2,180 items for every model of Table 2. Each column marks its best and runner-up; the shaded row is our main configuration.

Model Requests	key 786	meter 773	length 609	range 744	final 756	rhythm 761	leap 756	structure 733	All 2180
Zero-shot									
Qwen3-4B	0.242	0.109	0.016	0.293	0.278	0.332	0.336	0.057	0.068
Qwen3-8B	0.125	0.079	0.043	0.179	0.168	0.152	0.246	0.010	0.064
Qwen3-32B	0.767	0.219	0.148	0.628	0.868	0.585	0.574	0.150	0.204
Llama-3.1-70B	0.719	0.234	0.097	0.671	0.837	0.608	0.640	0.117	0.197
MiniMax-M3	0.468	0.248	0.281	0.456	0.460	0.250	0.452	0.030	0.215
ChatMusician-7B	0.159	0.378	0.049	0.384	0.194	0.322	0.565	0.000	0.084
Trained (Qwen3-4B-Instruct-2507)									
SFT	0.660	0.724	0.819	0.777	0.333	0.603	0.683	0.610	0.346
RFT	0.617	0.464	0.256	0.761	0.660	0.812	0.737	0.538	0.342
Direct GRPO ($R = A$)	0.719	0.820	0.565	**0.907**	0.788	**0.937**	0.819	0.644	0.545
MusicRLVR (SFT init.)	0.670	**0.872**	**0.844**	0.867	0.487	0.629	0.798	0.506	0.424
MusicRLVR (Direct)	0.735	0.820	0.657	0.867	0.796	0.928	**0.853**	**0.658**	**0.580**
Trained (Qwen3-8B)									
MusicRLVR (Direct)	**0.805**	0.841	0.731	0.817	**0.905**	0.913	0.792	0.307	0.530

family it requests. The first row counts the items requesting each family.

D.2 Instruction Wording

In this section, we test robustness to instruction wording. Rendering the same constraint values in different words costs the matched Direct arm 0.122 joint satisfaction, against 0.050 for the untrained base model. Drawing each training prompt from three renderings removes the effect (-0.018, $p = 0.311$) and costs 0.057 on Mixed, so the wording sensitivity comes from the training distribution.

Every row pairs item for item with its own original, so the exact McNemar test applies; the constraint values are identical in both members of a pair and only the surface form changes. Paraphrase re-renders the 400 single and seen-combination problems in a third English wording, never used in training. Chinese re-renders the 300 Mixed items, and no arm is trained on the wording used there.

Three observations. First, the paraphrase effect scales with how much a run was trained on one wording: the matched Direct arm, the highest on the original wording, loses 0.122 under paraphrase, and the untrained model loses 0.050. Second, wording variation in training buys invariance at a price (Table 25). The English-augmented arm loses 0.05 to 0.11 on Single, Seen, High-k and Structure, whose items contain no held-out pair, and stays within 0.005 of the matched Direct arm on Unseen and Dense, where every item contains one; the multilingual arm loses on every suite. Third, adding a Chinese rendering to training does not improve Chinese compliance: the multilingual arm reaches 0.543 against the English-only augmented arm's 0.587 and the matched Direct arm's 0.553, with every completion clearing the parse gate. What transfers across languages is wording variety rather than the language itself, and the bottleneck is wording generalisation within a language.

D.3 Free-Form Requests

In this section, we test requests worded as a user would write them, whereas every MusicConstraintBench prompt lists its constraints in template sentences. We hand-write 60 requests (12, 16, 16 and 16 at $k = 1$ to 4) that state each constraint in free-form English, with varied order, synonyms (measures, time signature, lowest and highest note) and indirect phrasing ("bring it home on C", "no anacrusis"), for example "I'd like something in waltz time, so 3/4, and please make sure every measure adds up exactly." Parameter values stay inside the default sampler domain and no request contains a held-out pair, so the set changes only the wording. Each request keeps the format rules of the template, which the validation gate enforces for every model, and has a twin that renders the same constraints with the template. Both sets are scored by the unchanged verifiers under the protocol of Table 2 (`scripts/make_bench_freeform.py`; `paper/make_robustness_tables.py`).

Table 26 reports every model we ran. Direct MusicRLVR has the highest free-form rate among the 4B models, 0.583, and item by item solves more free-form requests than each of them: 11 to 3 against Direct GRPO ($R = A$), 21 to 9 against the SFT-initialised run, 18 to 8 against RFT, 21 to 5 against SFT and 32 to 1 against Qwen3-4B. Every trained model clears the validation gate on at least

Table 24: Instruction-wording robustness on the matched runs. "Lost" and "gained" count items that flip in each direction. The wording-augmented arm draws each training prompt from three English renderings; the multilingual arm adds a fourth, Chinese one. Qwen3-8B's Chinese figure is set by the parse gate: 0.74 of its completions there omit the `X:` reference number, which no other arm does.

Model	Variant	Original	Variant	Lost	Gained	p
Qwen3-4B zero-shot	Paraphrase	0.163	0.113	49	29	0.031
SFT (4B)	Paraphrase	0.580	0.540	40	24	0.060
Direct GRPO (R=A)	Paraphrase	0.853	0.805	29	10	0.0034
MusicRLVR (Direct)	Paraphrase	0.900	0.777	54	5	1.9×10^{-11}
MusicRLVR (Qwen3-8B)	Paraphrase	0.810	0.787	27	18	0.233
MusicRLVR wording-aug.	Paraphrase	0.815	0.797	21	14	0.311
MusicRLVR multiling.-aug.	Paraphrase	0.797	0.838	13	29	0.020
Qwen3-4B zero-shot	Chinese	0.160	0.013	46	2	8.4×10^{-12}
SFT (4B)	Chinese	0.477	0.377	52	22	0.00064
Direct GRPO (R=A)	Chinese	0.797	0.447	111	6	4.0×10^{-26}
MusicRLVR (Direct)	Chinese	0.807	0.553	78	2	5.4×10^{-21}
MusicRLVR (Qwen3-8B)	Chinese	0.657	0.187	143	2	4.8×10^{-40}
MusicRLVR wording-aug.	Chinese	0.740	0.587	50	4	3.8×10^{-11}
MusicRLVR multiling.-aug.	Chinese	0.737	0.543	74	16	4.4×10^{-10}

Table 25: What wording augmentation costs, against the matched Direct run under the same recipe, seed, budget and group size. Both augmented arms spend the same 300 updates over more renderings.

Suite	MusicRLVR (Direct)	wording-aug.	multiling.-aug.
Mixed	0.807	0.740	0.737
Single	0.985	0.915	0.915
Seen	0.815	0.715	0.680
Unseen	0.595	0.600	0.500
High-k	0.575	0.463	0.506
Structure	0.967	0.917	0.667
Dense (k=4)	0.362	0.360	0.294
Edge	0.310	0.292	0.270
Paraphrase	0.777	0.797	0.838
Chinese	0.553	0.587	0.543

58 of the 60 free-form requests, against 24 and 26 for the untrained Qwen3-4B and Qwen3-8B. The Qwen3-8B policy reaches 0.717, the highest rate overall.

D.4 General Capability

In this section, we evaluate both Direct policies and their base models on three general benchmarks: IFBench (Pyatkin et al., 2026), all 300 test prompts scored by the official strict and loose checkers at the prompt level; MMLU-Redux (Gema et al., 2025), a fixed subject-proportional sample of 1,000 questions answered zero-shot; and GSM8K (Cobbe et al., 2021), the full 1,319-question test split with zero-shot chain of thought. Decoding is greedy under each model's own chat template with at most 4,096 new tokens, and Qwen3-8B runs with thinking disabled, as in Table 2. Table 27 gives the item-level comparison behind Table 4; no difference is significant at either scale (exact McNemar, $p \geq 0.096$).

D.5 Paraphrase and Training Dynamics

In Table 28, we re-render the single and seen-combination problems with alternative wordings of identical constraint values, and in Table 29, we give the numeric values behind Figure 7(d).

D.6 Rejection Sampling and a Music-Specialised Baseline

In Table 55(b), we spend the verifier at inference time instead: for each item we draw eight completions, matching the GRPO group size $G = 8$ and the rollout temperature (1.0), from both trained models, so selection sees the same per-prompt candidate count as one GRPO group, and report pass@k under the unbiased estimator $(1 - \binom{n-c}{k}/\binom{n}{k})$ for c of n samples satisfying all constraints, averaged over items). The results show that selection helps both models and MusicRLVR stays ahead at every matched candidate budget, on both suites, so verifier-guided selection stacks with training rather than substituting for it; note that, unlike the trained policy alone, best-of-N requires the verifier and N generations and N verifications at deployment.

ChatMusician-7B (Yuan et al., 2024) is evaluated zero-shot under the same protocol as every other untrained model, through its released `Human:/Assistant:` template and greedy decoding (context 2,048 tokens). Under A, it reaches 0.160 on the mixed suite and 0.000 on the structure split, where its parse rate is 0.158. Under the additional format gate its score is zero on all eight suites (Tables 16 and 18), mainly because required headers are missing. Trained on corpus ABC, it writes repeat signs on most form prompts, which

Table 26: Free-form requests (60 items) against their template-worded twins, which request the same constraints. "Free-form gate" counts completions that pass the validation gate; "lost" and "gained" count items that flip from the template twin to the free-form request; p is the exact McNemar test on these flips.

Model	Template	Free-form	Free-form gate	Lost	Gained	p
Qwen3-4B (zero-shot)	0.117	0.067	24/60	6	3	0.51
SFT	0.483	0.317	60/60	12	2	0.013
RFT	0.500	0.417	58/60	7	2	0.18
Direct GRPO ($R = A$)	0.700	0.450	60/60	16	1	< 0.001
MusicRLVR (SFT init.)	0.617	0.383	60/60	15	1	< 0.001
MusicRLVR (Direct)	0.717	0.583	60/60	10	2	0.039
Qwen3-8B (zero-shot)	0.150	0.133	26/60	4	3	1.00
MusicRLVR (Direct, Qwen3-8B)	0.767	0.717	60/60	8	5	0.58

Table 27: General benchmarks before and after training (accuracy, %). MusicRLVR is the Direct policy at each scale; "only ours / base" counts items that only one of the two models answers correctly, and p is the exact McNemar test on them.

	Qwen3-4B				Qwen3-8B			
Benchmark	Base	MusicRLVR	Only ours / base	p	Base	MusicRLVR	Only ours / base	p
IFBench, strict ($n = 300$)	32.0	29.3	5 / 13	0.096	25.7	25.3	4 / 5	1.00
IFBench, loose ($n = 300$)	33.3	31.7	9 / 14	0.40	30.3	30.0	5 / 6	1.00
MMLU-Redux ($n = 1{,}000$)	83.9	84.4	14 / 9	0.40	81.5	82.6	25 / 14	0.11
GSM8K ($n = 1{,}319$)	93.9	94.1	15 / 13	0.85	93.0	93.3	16 / 12	0.57

Table 28: Diagnostic study: instruction paraphrase, scored by A (the format audit leaves these verdicts unchanged). The original column averages the single and seen-combination splits; the paraphrased column re-renders the same 400 problems with alternative wordings of identical constraint values. Rows are the diagnostic lineage; Table 24 reports the matched runs.

Model	Original wording	Paraphrased	Δ
SFT	0.580	0.535	-0.045
MusicRLVR (diag.)	0.735	0.688	-0.047

the instruction forbids and the parse gate rejects. Its training objective is stylistic generation rather than stated requirements.

D.7 Rejection-Sampling Fine-Tuning

In this section, we use rejection-sampling fine-tuning (RFT) to ask whether the verifier's signal is better spent filtering supervised targets than as a reward. For each of the first 1,800 prompts of the reinforcement pool we draw 8 completions from Qwen3-4B-Instruct-2507 at temperature 1.0, 14,400 in all, the number GRPO samples over 300 updates of 48. For each prompt we keep the first completion that passes the gate and satisfies every requested constraint, which leaves 773 pairs, and fine-tune the full model on them for three epochs (learning rate 10^{-5}, effective batch 16, 147 steps, seed 43). Evaluation follows the protocol of Table 2.

D.8 Training versus Inference-Time Selection

In this section, we compare training with inference-time selection. For each Mixed item we draw 64 completions at temperature 1.0 and report pass@k under the unbiased estimator of Appendix D.6; with the verifier as the selector, pass@k is the accuracy of best-of-k reranking. The results in Table 30 show that, at every budget, the trained policy outscores its base model reranked from the same number of samples. To match the trained policy's single greedy decode, Qwen3-4B needs more than 64 reranked samples and Qwen3-8B between 32 and 64.

Table 3 extends the comparison to all eight suites at a budget of eight samples per item. Mixed reuses the 64 samples above under the same estimator; the other seven suites draw eight completions per item at temperature 1.0 (`scripts/run_rerank8_card0.sh` and `run_rerank8_card4.sh`, summarised by `src/analyse_rerank8.py`). Reranking raises the trained policy on every suite, by 0.008 to 0.262, and on every suite the untrained model's best of eight stays below the trained policy's single greedy decode.

D.9 Constraint-Count Ladder

In this section, we extend the number of constraints beyond the training range, as training prompts use $k \in \{1, 2, 3\}$. `src/gen_k_ladder.py` draws suites of 300 items at $k = 2$, 3, 5, 6 and 7 with the sampler and satisfiability filter of Dense, which is the $k = 4$ rung. The four held-out pairs form a perfect matching over the eight families, so any set of five or more families contains one; the ladder therefore keeps a held-out pair in every item at every rung, as Dense does, and varies only k. Because

Table 29: Diagnostic study: numeric values behind Figure 7(d). Curve rows come from a full-retention re-run of the SFT-initialised reference configuration; "original run" gives the three checkpoints the original run retained. Its step-300 value 0.660 is a separate evaluation of the weights Table 45 scores at 0.663, within the re-evaluation shift of Appendix B.10; the two are reported separately. The last row uses the symmetric structure reward of Appendix E.2; it and the symmetric row of Table 45 (0.590 on Mixed) are that reward's two separate runs.

Step	50	100	150	200	250	300
All satisfied	0.570	0.593	0.633	0.687	0.693	0.703
`meter`	0.716	0.852	0.901	0.926	0.914	0.926
`final`	0.505	0.505	0.558	0.642	0.632	0.632
`key`	0.838	0.838	0.787	0.838	0.800	0.812
`structure`	0.767	0.836	0.808	0.740	0.822	0.781
All satisfied (original run, retained ckpts)	–	–	–	0.630	0.640	0.660
All satisfied (symmetric structure reward)	0.477	0.463	0.463	0.530	0.537	0.553

Table 30: Mixed suite: greedy all-satisfied rate and pass@k from 64 samples at temperature 1.0.

Model	Greedy	pass@1	pass@8	pass@64
Qwen3-4B zero-shot	0.160	0.132	0.393	0.637
SFT (4B)	0.477	0.417	0.847	0.973
RFT (4B)	0.543	0.499	0.740	0.867
Direct GRPO ($R = A$)	0.797	0.765	0.878	0.920
MusicRLVR (SFT init.)	0.577	0.653	0.946	1.000
MusicRLVR (Direct)	0.807	0.778	0.897	0.937
Qwen3-8B zero-shot	0.123	0.135	0.398	0.713
MusicRLVR (Direct, 8B)	0.657	0.667	0.767	0.823

`length` and `structure` cannot both be requested, $k = 7$ is the largest admissible value. Table 31 lists every arm evaluated on the ladder.

Table 31: All-satisfied rate on the constraint-count ladder ($k = 4$ is Dense). Every item contains a held-out pair; training prompts hold at most three constraints.

Model	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k = 7$
Qwen3-4B (zero-shot)	0.073	0.037	0.008	0.003	0.000	0.000
SFT	0.467	0.357	0.290	0.140	0.117	0.073
RFT	0.530	0.207	0.098	0.053	0.007	0.003
Direct GRPO ($R = A$)	0.710	0.487	0.322	0.170	0.060	0.063
MusicRLVR (SFT init.)	0.630	0.403	0.366	0.187	0.200	0.117
MusicRLVR (Direct)	0.647	0.530	0.362	0.230	0.160	0.087
MusicRLVR (Direct, Qwen3-8B)	0.707	0.493	0.404	0.273	0.250	0.153
MusicRLVR (Direct, wording-aug.)	0.680	0.477	0.360	0.167	0.117	0.087

D.10 Direct Reward Ablations

In Table 6, we vary one part of Equation 5 at a time from the Direct configuration of Table 2. Every arm uses seed 43, 300 updates of 48 completions, the same rollout server and the evaluation of Table 2; at the first update every arm except the 21% pool samples exactly the default's rollouts. The arms are:

- **Gate on A only.** An output that the gate rejects for repeat signs or for a degenerate or polyphonic score, but that music21 still parses, receives $\lambda \frac{1}{k} \sum_{(f,\theta)\in C} s_f$ instead of 0. A keeps the gate, so the joint term and the evaluation criterion are unchanged; outputs with no ABC block or no parse still score 0.
- **$\lambda = 0$, 0.5 and 1.** The outer mixture of Equation 5; $\lambda = 0$ is Direct GRPO ($R = A$) of Table 2.
- **Binary form credit.** $s_{\texttt{structure}} = v_{\texttt{structure}}$.
- **Symmetric form credit** and **equal form weights.** $0.5\bar{e} + 0.5\bar{d}$, with credit 1 for a passing form and 0 for four identical units, and $0.5\bar{e} + 0.5v_{\text{contrast}}$ (Appendix E.2).
- **Form share 21%.** The prompt pool before upsampling: 1,282 of 6,000 prompts contain `structure`, and $k = 1/2/3$ is split 1,863/3,076/1,061 against 2,046/3,060/894.

Table 32 gives the counts behind the All column of Table 6, with exact McNemar tests against the default, Holm-corrected over the eight arms. Without the all-satisfied bonus ($\lambda = 1$) the policy fails to parse 25 outputs: 19 repeat notes without bar lines until the token limit and 6 end before four bars.

D.11 Edge Subsets and a Stricter Key Criterion

In this section, we rescore Edge subsets and apply a stricter key criterion; both analyses use saved per-item verdicts, without regeneration or retraining. `src/rescore_subsets.py` computes the key-sensitivity audit under the historical gate; the final `make_checkpoint_assets.py` pass uses the same A verdicts for Edge subsets.

Edge mixes two factors. Table 34 splits it by whether an item requests one of the four held-out pairs. On the 329 items without a held-out family pair, the diagnostic reference reaches 0.210 against 0.100 for the warm start (exact McNemar $p = 2.03 \times 10^{-6}$) and 0.195 for MiniMax-M3 ($p = 0.685$), a wider margin over the warm start than on Edge as a whole.

Table 33 instead splits Edge by the factor it extrapolates: items that request a meter outside $\{4/4, 3/4, 6/8, 2/4\}$ and items that request a length outside $\{8, 12, 16\}$ bars, for every model of Table 2.

The deployed key check requires the header to

Table 32: Direct reward ablations over all 2,180 items: parsed outputs, solved items, and the items solved only by the variant or only by the default, with exact McNemar p and its Holm adjustment over the eight arms.

Variant	Parsed	Solved	Rate	Only variant	Only default	p	Holm p
Default	2176	1265	0.580	–	–	–	–
Gate on A only	2179	1164	0.534	125	226	7.7×10^{-8}	3.9×10^{-7}
$\lambda = 0$ ($R = A$)	2180	1188	0.545	152	229	0.0001	0.0002
$\lambda = 0.5$	2168	1095	0.502	150	320	3.2×10^{-15}	2.6×10^{-14}
$\lambda = 1$ (no bonus)	2155	1206	0.553	146	205	0.0019	0.0019
Binary form credit	2180	1190	0.546	129	204	4.7×10^{-5}	0.0002
Symmetric form credit	2180	1122	0.515	134	277	1.5×10^{-12}	1.0×10^{-11}
Equal form weights	2178	1186	0.544	142	221	4.0×10^{-5}	1.6×10^{-4}
Form share 21%	2179	1130	0.518	188	323	2.5×10^{-9}	1.5×10^{-8}

Table 33: Edge all-satisfied rate on the items whose requested meter or length lies outside the RL prompt domain.

Model	Unseen meter ($n = 160$)	Unseen length ($n = 127$)	All Edge
Qwen3-4B	0.000	0.000	0.010
Qwen3-8B	0.019	0.024	0.032
Qwen3-32B	0.037	0.063	0.112
Llama-3.1-70B	0.019	0.055	0.058
MiniMax-M3	0.138	0.142	0.160
ChatMusician-7B	0.044	0.008	0.046
SFT	0.144	0.055	0.076
RFT	0.131	0.102	0.128
Direct GRPO ($R = A$)	0.269	0.189	0.300
MusicRLVR (SFT init.)	0.188	0.094	0.134
MusicRLVR (Direct)	0.287	0.181	0.310
MusicRLVR (Direct, Qwen3-8B)	0.300	0.260	0.252

match and the analyzed key to be the requested key or its relative (§C.3). Tables 35 and 36 hold the historical format gate fixed and report every rate with that tolerance and without it, that is with `key` satisfied only when header and analysis both give the requested key. Rates stay unchanged or decrease. The largest baseline decrease is 0.060 (Qwen3-32B on Unseen); the largest decrease for the fixed MusicRLVR checkpoint is 0.030 on Seen. The diagnostic MusicRLVR checkpoint remains the highest-scoring model on seven of the eight splits.

D.12 Significance Tests

In this section, we report significance tests for the diagnostic reward-design comparisons; the matched Direct-versus-SFT-initialised tests are in Appendix D.13.

Table 37 reports exact McNemar tests for the single-run diagnostic reward-design experiment under the main-table success criterion, pairing the same item IDs; Appendix D.16 lists the success counts underlying the historical reference rates. Table 38 reports the Edge comparison.

D.13 Matched Comparison

In this section, we detail the matched comparison, which supplies the three 4B RL rows in Table 2. All runs use seed 43, checkpoint 300, the same 6,000-prompt training pool and 48 completions per update. Two policy accelerators use micro-batch 2 and accumulation 12; a third serves rollouts with memory fraction 0.92. The remaining GRPO settings match Table 7. Direct binary uses $\lambda = 0$ ($R = A$), and both MusicRLVR routes use $\lambda = 0.7$. The reward weight is the only changed training setting between the two Direct arms. The SFT-initialised arm changes both the initial policy and its fixed KL reference. All evaluations use the same item IDs and the all-satisfied criterion A.

Pooled comparison. The eight suites are disjoint item sets whose union is the 2,180 items of MusicConstraintBench, so the same paired test applies to the union. Against binary reward, Direct MusicRLVR satisfies 1257 items against 1196 (0.577 against 0.549), winning 221 and losing 160 of the 381 items on which the two disagree (exact two-sided $p = 0.0021$). Against the SFT-initialised route it satisfies 1257 against 924, winning 549 and losing 216 ($p < 10^{-4}$); the binary-reward arm also leads that route, 1196 against 924 ($p < 10^{-4}$).

Per-suite breakdown. Graded credit has the higher rate on all eight of the suites, and the pooled comparison above is significant ($p = 0.0021$). Against the SFT-initialised route, Direct is ahead on seven suites, six of them significant after the eight-suite Holm correction of Table 42. All three models pass the parse gate on every original-suite item; on Edge the Direct MusicRLVR model passes 496/500, while the binary-reward and SFT-initialised models pass 500/500 (Tables 40 and 41).

What the residual failures look like. Table 43 decomposes the 58 Mixed items the Direct MusicRLVR checkpoint does not satisfy, from the same saved per-item verdicts as Table 39. The failures are shallow rather than pervasive: 48 of the 58 miss a single family and only 2 miss three, and none

Table 34: Diagnostic study: Edge split by held-out-pair membership, using the main-table success criterion. Pair-free items contain none of the four held-out family pairs.

Model	All ($n = 500$)	Pair-free ($n = 329$)	With held-out pair ($n = 171$)
Qwen3-4B	0.010	0.012	0.006
Qwen3-8B	0.032	0.049	0.000
Qwen3-32B	0.112	0.125	0.088
Llama-3.1-70B	0.058	0.067	0.041
MiniMax-M3	0.160	0.195	0.094
ChatMusician-7B	0.046	0.055	0.029
SFT	0.076	0.100	0.029
MusicRLVR (re-evaluated)	0.162	0.210	0.070

Table 35: Diagnostic study: All-satisfied rate with the relative-key tolerance and without it (hybrid/exact key), six original suites, holding the historical format gate fixed. Baselines show the paired rates; the two MusicRLVR rows report the fixed-checkpoint rate under each key rule.

Model / key rule	Mixed	Single	Seen	Unseen	High-k	Structure
Qwen3-4B	0.160/0.157	0.225/0.210	0.100/0.090	0.040/0.030	0.037/0.031	0.100/0.100
Qwen3-8B	0.123/0.117	0.210/0.200	0.115/0.110	0.050/0.050	0.006/0.006	0.000/0.000
Qwen3-32B	0.363/0.340	0.535/0.485	0.250/0.240	0.190/0.130	0.094/0.062	0.267/0.267
Llama-3.1-70B	0.380/0.370	0.695/0.675	0.300/0.295	0.195/0.185	0.056/0.056	0.167/0.167
MiniMax-M3	0.367/0.363	0.580/0.550	0.315/0.310	0.190/0.185	0.100/0.100	0.008/0.008
ChatMusician-7B	0.160/0.147	0.290/0.280	0.130/0.120	0.090/0.085	0.006/0.000	0.000/0.000
SFT	0.477/0.453	0.745/0.715	0.415/0.385	0.440/0.415	0.231/0.225	0.600/0.592
MusicRLVR (hybrid)	0.663	0.835	0.635	0.595	0.356	0.808
MusicRLVR (strict key)	0.637	0.815	0.605	0.580	0.344	0.783

Table 36: Diagnostic study: Hybrid/exact key on Dense and Edge, holding the historical format gate fixed; MusicRLVR uses the single re-evaluation.

Model	Dense	Edge
Qwen3-4B	0.008/0.006	0.010/0.008
Qwen3-8B	0.020/0.018	0.032/0.032
Qwen3-32B	0.074/0.056	0.112/0.102
Llama-3.1-70B	0.040/0.032	0.058/0.048
MiniMax-M3	0.088/0.084	0.160/0.152
ChatMusician-7B	0.020/0.014	0.046/0.042
SFT	0.290/0.266	0.076/0.070
MusicRLVR (re-evaluated)	0.370/0.342	0.162/0.150

fails the parse gate. `structure` has the largest per-family deficit (22 of the 73 items requesting it) and the top sole miss (18 items); `length` is next in missed count (14) and `key` and `final` each miss 11. The split-level patterns of Appendix E.1 come from the diagnostic checkpoint.

D.14 Diagnostic Reference Results

In Table 44, we give the diagnostic evaluations. Unless a diagnostic explicitly states another setting, its RL results refer to the SFT-initialised checkpoint in this table. The Direct route is reported only from the matched runs of Table 2, so no Direct row appears here.

D.15 Diagnostic Reward-Design Ablations

In Table 45, we run the reward-design comparison from the SFT checkpoint of the diagnostic study. The results show that raising form exposure from 21% to 33% takes Structure from 0.250 to 0.525, and graded form credit then takes it to 0.808 while the other five suites move by at most 0.025; grading contrast as well (symmetric) drops it to 0.200. Figure 7 traces these runs: under graded credit Structure outputs pass repetition on 0.99 and contrast on 0.87 of items (c), and every learning rate, KL coefficient and group size tested keeps Mixed above the SFT start, 0.540–0.837 against 0.477 (b). In the Direct configuration (Table 6) Structure stays between 0.867 and 0.967 under every variant, and every variant loses most on High-k.

D.16 Fixed-Checkpoint Evaluation

In this section, we report success counts for the diagnostic reference checkpoint, `outputs/grpo_4b_v2/checkpoint-300`. Table 46 gives its counts on the six original suites; on Dense and Edge it reaches $185/500$ and $81/500$. Its first evaluation supplies the original-suite rates; the separate re-evaluation is documented in Appendix B.10. Success counts for the three matched checkpoints are in Table 39.

D.17 Hyper-parameter Sweeps

In Table 47, we list the values plotted in Figure 7(b). Each cell is one training run per setting with a single parameter moved from the SFT-initialised reference configuration, scored on Mixed and Structure. The reward mixing weight is swept the same way in Appendix D.18.

Stability across settings. Every setting exceeds the SFT Mixed baseline of 0.477. Learning rate

Table 37: Diagnostic study: Single-run reward-design comparisons (nominal exact McNemar p) over per-item outcomes.

Comparison	Mixed	Single	Seen	Unseen	High-k	Struct.
Reference run vs. SFT	$< 10^{-4}$	0.002	$< 10^{-4}$	$< 10^{-4}$	0.007	$< 10^{-4}$
Reference run vs. binary credit, 21% share	0.076	0.597	1.000	0.312	1.000	$< 10^{-4}$
Reference run vs. binary credit, 33% share	0.822	0.583	0.625	0.771	0.678	$< 10^{-4}$

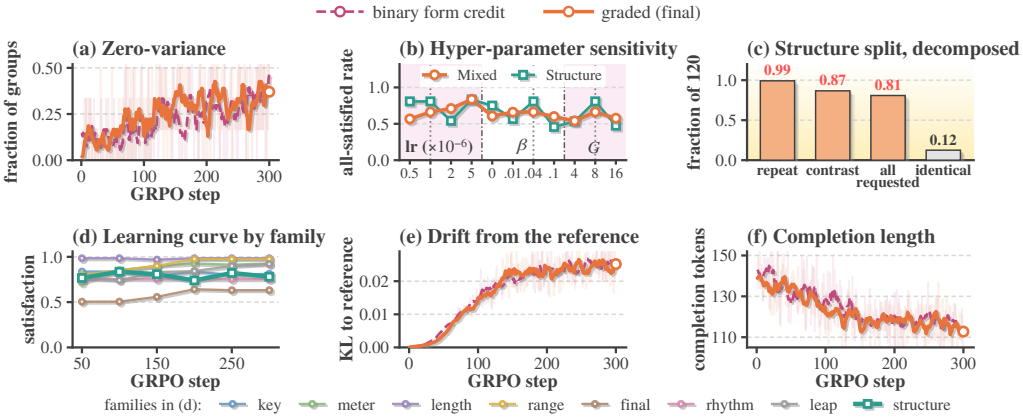


Figure 7: **Reward-design ablations (diagnostic study). (a)** Zero-variance groups. **(b)** Sensitivity to learning rate, β and G (dotted: reference). **(c)** Form verifier decomposed. **(d)** Per-family satisfaction. **(e–f)** KL and length.

Table 38: Diagnostic study: Paired comparison on Edge. "Only RL" and "Only baseline" count discordant items in the designated single evaluation of each model.

Comparison	Only RL	Only baseline	Exact p	Direction
Edge: RL vs. MiniMax-M3	62	61	1.000	No detected difference

moves Mixed most (0.270), against 0.063 for the KL coefficient and 0.123 for group size.

The G sweep fixes 48 completions per optimiser step. Accordingly $G = 4, 8, 16$ uses $12, 6, 3$ prompt groups per step, for 3,600, 1,800, 900 prompt draws in 300 steps. The corresponding logged epochs are $0.60, 0.30, 0.15$. The sweep therefore compares allocations of a fixed completion budget, with prompt exposure falling as G grows.

D.18 Reward Mixing Weight

In Table 48, we move λ of Equation 5, which weights the per-family average against the binary all-satisfied term, while every other setting stays at the SFT-initialised reference configuration. It is reported separately from Table 47 because that table lists the values plotted in Figure 7(b), and because the form decomposition below is what makes this sweep interpretable.

The default scores highest on both suites among the three weights tested. Against $\lambda = 0.7$, the low arm's Mixed comparison gives exact McNemar $p = 0.368$, while its Structure score is lower ($p = 3.54 \times 10^{-9}$); the high arm scores lower on both ($p = 0.002$ and $p = 4.74 \times 10^{-15}$). Table 6 varies λ over 0, 0.5, 0.7 and 1 in the Direct configuration, where 0.7 is likewise highest over all 2,180 items.

Contrast distinguishes the tested weights. The last two columns separate the halves of the form constraint. Repetition is satisfied on 119–120 of the 120 items at every weight, while contrast passes 55 items at $\lambda = 0.5$ and 45 at $\lambda = 1.0$ against 104 at the default: the decrease on Structure accompanies a drop in contrast, not in repetition. The all-satisfied counts behind those rates are 51, 97 and 39; each deficit relative to the contrast count consists of items passing the form check and missing a co-requested family. Outputs whose four units are all identical – which satisfy every required-equal pair and no required-distinct one – rise from 15 items at the default to 65 and 74 at the two ends. The same pattern holds across the rows of Table 45: repetition passes 119–120 items, and the binary form pass count equals the contrast pass count.

Where the weight can act. Write m_i for the per-family average in Equation 5, setting it to zero for a gate-rejected rollout. For $\lambda > 0$, a group in which every rollout has $A(y_i, C) = 0$ has $R(y_i, C) = \lambda m_i$. Equation 4 then gives $\hat{A}_i = (m_i - \bar{m})/(\sigma_m + \epsilon/\lambda)$, where σ_m is the group sample standard deviation of m: the scaling cancels except through the stabiliser. The mixture also controls the reward gap between full and partial success. Consider two gate-passing rollouts for the same prompt: one satisfies all k families; the other fails exactly one family, retaining graded

Table 39: Success counts and rates for the matched final checkpoints.

Split	Direct, binary	Direct, MusicRLVR	SFT init., MusicRLVR
Mixed	239/300 (0.797)	242/300 (0.807)	173/300 (0.577)
Single	192/200 (0.960)	197/200 (0.985)	176/200 (0.880)
Seen	149/200 (0.745)	163/200 (0.815)	117/200 (0.585)
Unseen	113/200 (0.565)	119/200 (0.595)	106/200 (0.530)
High-k	75/160 (0.469)	92/160 (0.575)	56/160 (0.350)
Structure	109/120 (0.908)	116/120 (0.967)	46/120 (0.383)
Dense	161/500 (0.322)	181/500 (0.362)	183/500 (0.366)
Edge	150/500 (0.300)	155/500 (0.310)	67/500 (0.134)

Table 40: Parse-gate pass rates on the six original suites, for every model of Table 2. SFT and every GRPO-trained checkpoint clear the gate on every item.

Model	Mixed	Single	Seen	Unseen	High-k	Structure
Qwen3-4B	0.513	0.450	0.510	0.545	0.362	0.925
Qwen3-8B	0.367	0.465	0.345	0.305	0.219	0.158
Qwen3-32B	0.973	1.000	0.960	0.945	0.938	0.983
Llama-3.1-70B	0.970	1.000	0.980	0.980	0.944	1.000
MiniMax-M3	0.577	0.670	0.565	0.465	0.531	0.083
ChatMusician-7B	0.693	0.800	0.680	0.670	0.575	0.158
SFT (4B)	1.000	1.000	1.000	1.000	1.000	1.000
Direct GRPO ($R = A$)	1.000	1.000	1.000	1.000	1.000	1.000
MusicRLVR (SFT init.)	1.000	1.000	1.000	1.000	1.000	1.000
MusicRLVR (Direct)	1.000	1.000	1.000	1.000	1.000	1.000
RFT (4B)	1.000	0.980	0.995	1.000	0.988	1.000
MusicRLVR (Direct, Qwen3-8B)	1.000	1.000	1.000	1.000	1.000	1.000

Table 41: Parse-gate pass rates for the matched evaluations.

Model	Mixed	Single	Seen	Unseen	High-k	Structure	Dense	Edge
Direct GRPO ($R = A$)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
MusicRLVR (SFT init.)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
MusicRLVR (Direct)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992

credit $s \in [0, 1)$ for that family. Their reward gap is

$$\Delta R = (1 - \lambda) + \frac{\lambda(1 - s)}{k},$$

which is $(1-s)/k$ at $\lambda = 1$ and $0.3+0.7(1-s)/k$ at the default. The binary term thus adds an explicit bonus for joint satisfaction; its effect on normalised advantages also depends on the other rewards in the group. The default mixture gives the highest compliance and the fewest all-identical outputs.

D.19 Per-Family Satisfaction on the Original Suites

In this section, we break the all-satisfied rate down by family. Table 44 reports the all-satisfied rate, which is zero whenever any single constraint fails; Table 50 reports each family's satisfaction rate on each split for the same checkpoint, and Figure 8 gives the same per-family view on Mixed across models.

The mixed suite includes 73 structure-family items. Their satisfaction rate is 0.644 for MusicRLVR against 0.726 for SFT. On the dedicated structure split, the all-satisfied rate is 0.808 against 0.600; equal prompt weighting gives 0.637 against 0.522 (Appendix B.8). The splits differ in their requested constraints and prompt weights. `length` and `range` maintain high satisfaction across the splits.

D.20 Performance by Number of Constraints (k)

In Table 52, we break the mixed suite ($n = 300$; 100/100/60/40 items at $k = 1/2/3/4$) down by the number of constraints, using the same saved diagnostic reference evaluations as Table 44. Figure 1 instead pools all 2,180 items for the matched runs; Table 51 lists its values.

The endpoint ratios vary across models. Llama-3.1-70B falls from 0.720 at $k = 1$ to 0.050 at $k = 4$, a 14.4× drop; MiniMax-M3 falls 11.2× and the untrained 4B base 13.6×. The diagnostic reference rates are 0.850 and 0.475, giving the smallest endpoint ratio in the table (1.8×), against 2.4× for SFT. The matched by-k curves, pooled over all suites, are in Figure 1.

D.21 Variety and Degeneration of the Matched Checkpoints

In Tables 53 and 54, we repeat the within-tune statistics of Table 56 for every model of Table 2 over all 2,180 items (`src/audit_quality_main.py`). Pitch-class entropy uses the whole tune; the distinct-bar and distinct pitch 4-gram ratios use the first eight bars, so tunes of different lengths are compared on

Table 42: Exact two-sided McNemar tests for the matched Direct MusicRLVR checkpoint against each control; discordant counts are successes by only one checkpoint, and Holm correction is applied over the eight suites within each comparison.

Comparison	Split	Only Direct	Only control	Exact p	Holm p
Binary reward	Mixed	19	16	0.7359	1.0000
	Single	5	0	0.0625	0.3125
	Seen	25	11	0.0288	0.1729
	Unseen	21	15	0.4050	1.0000
	High-k	32	15	0.0186	0.1304
	Structure	7	0	0.0156	0.1250
	Dense	78	58	0.1029	0.4117
	Edge	42	37	0.6530	1.0000
SFT init.	Mixed	90	21	$< 10^{-4}$	$< 10^{-4}$
	Single	22	1	$< 10^{-4}$	$< 10^{-4}$
	Seen	65	19	$< 10^{-4}$	$< 10^{-4}$
	Unseen	36	23	0.1175	0.2350
	High-k	58	22	$< 10^{-4}$	0.0002
	Structure	71	1	$< 10^{-4}$	$< 10^{-4}$
	Dense	90	92	0.9409	0.9409
	Edge	122	34	$< 10^{-4}$	$< 10^{-4}$

Table 43: Matched study: the 58 unsatisfied Mixed items of the Direct MusicRLVR checkpoint. "Missed" counts items requesting a family whose verifier fails; "sole miss" restricts to the 48 items that miss exactly one family. `rhythm` is omitted, with no miss among the items requesting it.

Family	Requested	Missed	Sole miss
`final`	95	11	7
`structure`	73	22	18
`key`	80	11	9
`length`	65	14	10
`leap`	89	7	4
`meter`	81	3	0
`range`	80	2	0
Items by number of families missed: 1: 48, 2: 8, 3: 2; parse-gate failures: 0.			

Table 44: Diagnostic reference results (SFT-initialised route): all-satisfied rates above, parse-gate pass rates below. A different RL checkpoint from the matched comparison of Table 2. Colours rank rows within this table; missing entries were not evaluated.

	Six original splits						Later splits	
Model	Mixed	Single	Seen	Unseen	High-k	Structure	Dense	Edge
Zero-shot								
Qwen3-4B	0.160	0.225	0.100	0.040	0.037	0.100	0.008	0.010
Qwen3-8B	0.123	0.210	0.115	0.050	0.006	0.000	0.020	0.032
Qwen3-32B	0.363	0.535	0.250	0.190	0.094	0.267	0.074	0.112
Llama-3.1-70B	0.380	0.695	0.300	0.195	0.056	0.167	0.040	0.058
MiniMax-M3	0.367	0.580	0.315	0.190	0.100	0.008	0.088	0.160
ChatMusician-7B	0.160	0.290	0.130	0.090	0.006	0.000	0.020	0.046
Trained (4B)								
SFT (4B)	0.477	0.745	0.415	0.440	0.231	0.600	0.290	0.076
MusicRLVR (diag.)	**0.663**	**0.835**	**0.635**	**0.595**	**0.356**	**0.808**	**0.370**	**0.162**

Parse-gate pass rate (six original splits)						
Model	Mixed	Single	Seen	Unseen	High-k	Structure
Qwen3-4B	0.513	0.450	0.510	0.545	0.362	0.925
Qwen3-8B	0.367	0.465	0.345	0.305	0.219	0.158
Qwen3-32B	0.973	1.000	0.960	0.945	0.938	0.983
Llama-3.1-70B	0.970	1.000	0.980	0.980	0.944	1.000
MiniMax-M3	0.577	0.670	0.565	0.465	0.531	0.083
ChatMusician-7B	0.693	0.800	0.680	0.670	0.575	0.158
SFT (4B)	1.000	1.000	1.000	1.000	1.000	1.000
MusicRLVR (diag.)	1.000	1.000	1.000	1.000	1.000	1.000

equal spans. Three flags mark degenerate tunes: at most three distinct pitches, every bar identical, and more than half of consecutive notes repeating the previous pitch.

Against its base model, Direct MusicRLVR raises pitch-class entropy from 2.06 to 2.15, lowers the share of tunes with at most three pitches from 28.9% to 22.9% and of tunes dominated by

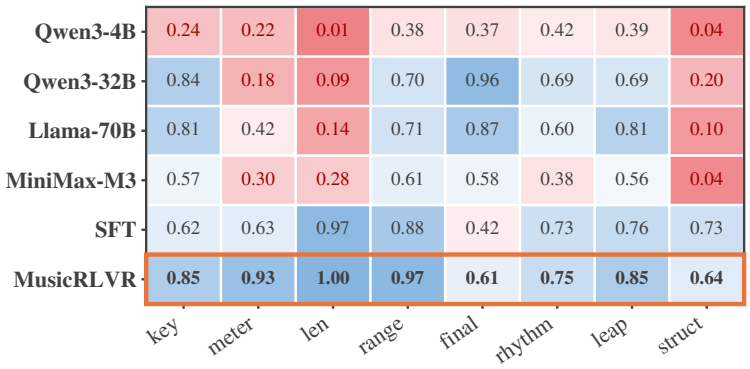


Figure 8: Diagnostic study: Per-family satisfaction on the mixed suite; MusicRLVR cells use the fixed checkpoint (Table 49); `structure` stays red for every untrained model.

Table 45: Reward-design ablations (diagnostic study): one 300-step run per RL row from the same SFT checkpoint. Rows give the credit for the form constraint and the share of training problems containing it (symmetric credit: Appendix E.2). Colours as in Table 2; the shaded row is the default reward.

Form credit	Mixed	Single	Seen	Unseen	High-k	Struct.
SFT, no GRPO	0.477	0.745	0.415	0.440	0.231	0.600
Binary, 21%	0.610	0.810	0.630	**0.635**	0.350	0.250
Binary, 33%	0.653	0.810	0.610	0.610	0.331	0.525
Graded, 33%	**0.663**	**0.835**	**0.635**	0.595	0.356	**0.808**
Symmetric, 33%	0.590	0.745	0.590	0.630	**0.388**	0.200

Table 46: Diagnostic study: success counts and all-satisfied rates for the SFT-initialised reference checkpoint on the six original suites, matching Table 44.

Split	SFT init. count	Rate
Mixed	199/300	0.663
Single	167/200	0.835
Seen	127/200	0.635
Unseen	119/200	0.595
High-k	57/160	0.356
Structure	97/120	0.808

Table 47: Diagnostic study: All-satisfied rate on the mixed and structure suites for each hyper-parameter setting, one training run each. * marks the SFT-initialised reference configuration.

Hyper-parameter	Value	Mixed	Structure
Learning rate	5×10^{-7}	0.567	0.808
	$1\times10^{-6\star}$	0.663	0.808
	2×10^{-6}	0.710	0.542
	5×10^{-6}	0.837	0.833
KL coefficient β	0	0.607	0.750
	0.01	0.660	0.567
	$0.04^\star$	0.663	0.808
	0.1	0.600	0.458
Group size G	4	0.540	0.533
	$8^\star$	0.663	0.808
	16	0.577	0.475

repeated notes from 8.5% to 1.2%. Table 54 keeps only the 332 items without a form constraint, which requests repetition by design, on which both models render at least eight bars. The same pattern holds there: entropy 2.02 to 2.17 and repeated-note tunes 12.3% to 2.1%.

Table 48: Diagnostic study: Reward mixing weight, one training run per setting, scored by the binary criterion of Equation 1. Repetition and contrast count the structure-split items passing all required-equal and all required-distinct unit-pair checks, respectively; * marks the default.

λ	Mixed	Structure	Repetition	Contrast
0.5	0.633	0.425	120/120	55/120
$0.7^\star$	0.663	0.808	119/120	104/120
1.0	0.570	0.325	119/120	45/120

Table 49: Diagnostic study: Per-family satisfaction on Mixed: the fraction of items requesting a family whose verifier passes. Δ is the fixed-checkpoint rate minus the SFT rate.

Family	Base	SFT	MusicRLVR	Δ
`key`	0.237	0.625	0.850	+0.225
`meter`	0.222	0.630	0.926	+0.296
`length`	0.015	0.969	1.000	+0.031
`range`	0.375	0.875	0.975	+0.100
`final`	0.368	0.421	0.611	+0.189
`rhythm`	0.416	0.727	0.753	+0.026
`leap`	0.393	0.764	0.854	+0.090
`structure`	0.041	0.726	0.644	-0.082

Table 50: Diagnostic study: per-family satisfaction rate for MusicRLVR across the splits. Each cell is the fraction of problems mentioning that family whose own verifier passes at the fixed checkpoint (other constraints may fail); families mentioned by fewer than five problems in a split are marked "–".

Family	Single	Seen	Unseen	High-k	Structure
`final`	0.783	0.636	0.407	0.635	–
`key`	0.862	0.887	0.565	0.671	0.643
`leap`	0.292	0.935	0.985	0.872	1.000
`length`	1.000	1.000	1.000	1.000	–
`meter`	0.812	0.903	0.897	0.890	1.000
`range`	1.000	0.982	0.955	0.980	1.000
`rhythm`	0.833	0.790	0.843	0.655	0.692
`structure`	1.000	0.510	0.667	0.554	0.867

Table 51: All-satisfied rate by number of constraints k over all eight suites (2,180 items), the values plotted in Figure 1. Each slice pools the suites containing that k; the trained rows are the matched runs of Table 2.

Model Items	$k=1$ 354	$k=2$ 614	$k=3$ 512	$k=4$ 700	k=1/k=4
Qwen3-4B	0.243	0.068	0.018	0.016	15.5×
Qwen3-32B	0.559	0.225	0.098	0.083	6.8×
Llama-3.1-70B	0.630	0.212	0.090	0.044	14.2×
MiniMax-M3	0.486	0.267	0.137	0.089	5.5×
SFT	0.723	0.349	0.178	0.277	2.6×
Direct GRPO ($R = A$)	0.977	0.661	0.357	0.361	2.7×
MusicRLVR (SFT init.)	0.757	0.430	0.283	0.353	2.1×
MusicRLVR (Direct)	0.992	0.673	0.406	0.419	2.4×

D.22 Length-Controlled Output Statistics

Table 55: **Diagnostic output analysis. (a)** Surface statistics (first 8 bars, 297 matched items). **(b)** Best-of-N (T=1.0, n=8, binary).

(a)

Model	Notes	PC ent.	Dist. bars	Dist. 4-gr.
SFT	42.7	1.930	0.417	0.419
MusicRLVR (diag.)	40.5	1.826	0.426	0.403

(b)

Suite	Model	p@1	p@2	p@4	p@8
Mixed	SFT	0.415	0.584	0.728	0.827
	MusicRLVR (diag.)	0.676	0.817	0.900	0.950
Struct.	SFT	0.692	0.861	0.938	0.967
	MusicRLVR (diag.)	0.844	0.953	0.978	0.983

In Table 55, we compare the first eight parsed bars of a common item set of the diagnostic study, because the full-tune ratios of Table 56 have length-dependent denominators. In this section, we give the uncontrolled statistics those ratios come from, and the cross-prompt duplication counts of Table 57. Both read historical-gate outputs, but they measure different things: the first are within-tune proxies, the second counts how often two different prompts produce the same score.

Table 56: Diagnostic study: distributional statistics of the single-run mixed-suite completions and 299 corpus tunes. No constraint names entropy or either distinctness measure; bar count is constrained on the 138/300 items requesting `length` or `structure`. Mean length differs between rows, so distinctness denominators are unmatched; Table 55 controls for that.

Model	Bars	PC ent.	Dist. bars	Dist. 4-gr.
Corpus	19.9	2.63	0.758	0.676
Base	29.3	2.06	0.366	0.391
SFT	39.2	2.00	0.276	0.294
RL, binary credit	32.2	1.85	0.284	0.289
MusicRLVR	28.7	1.89	0.313	0.292

Table 57: Diagnostic study: full-score duplication for the first eligible occurrence of each of 220 distinct prompts in that common set, identified by parsed event signatures and ignoring metadata, which measures duplication across prompts.

Model	Unique full scores	Most frequent score
SFT	167/220	7/220
MusicRLVR (diagnostic)	182/220	5/220

On the full completions, MusicRLVR raises distinct-bar fraction from 0.276 to 0.313 while mean length falls 39.2 → 28.7 bars. Under the length control of Table 55 the distinct-bar comparison is 0.417 → 0.426 on matched eight-bar prefixes. The matched checkpoints of Table 2 are analysed in Appendix D.21.

D.23 Output Length Statistics

In Table 58, we report the average number of bars and notes per parsed completion on the mixed suite, supplementing the quality metrics in Table 56. The supervised model over-generates (39.2 bars against 19.9 for the corpus), and the reinforcement stage pulls this back to 28.7 bars, consistent with a reward that favours concision without an unconditional length term, since a longer tune has more places to violate a constraint. The three larger zero-shot models sit near the corpus length (17.9–21.2 bars; the 4B base averages 29.3), and MiniMax-M3 parses on only 173 of 300 problems.

D.24 Zero-Variance Groups During Training

In this section, we track groups without a reward gradient during training. Equation 4 implies that a group whose G rollouts all receive the same reward contributes no reward gradient. The trainer logs the fraction of such groups per step (`frac_reward_zero_std`); Table 59 summarises it for the two reward variants of the main ablation.

The two diagnostic rows differ in both the reward and the share of structure prompts; the matched runs below change the reward alone.

The matched runs measure the same quantity on two arms that differ only in λ, holding the prompt mix fixed. Averaged over the first 50 updates, binary reward leaves 0.550 of groups with zero reward variance against 0.307 for the graded reward; over the last 50 the two reach 0.823 and 0.823, once most groups succeed outright, and over all 300 updates they average 0.717 and 0.660 (Figure 4(b)). The SFT-initialised arm runs lower throughout (0.120, 0.343, 0.245), so the fraction is compared between arms that share an initialisation.

Table 52: Diagnostic study: all-satisfied rate on the mixed suite by number of constraints k. The last column is the k=1 rate over the k=4 rate; a dagger marks curves not monotone in k (the k=3 and k=4 cells hold only 60 and 40 items, so one item moves them by 0.017 and 0.025).

Model	$k=1$	$k=2$	$k=3$	$k=4$	k=1/k=4
MusicRLVR (diag.)	0.850	0.640	0.517	0.475	1.8×
SFT (4B)	0.720	0.430	0.267	0.300	2.4×†
Qwen3-32B	0.650	0.300	0.133	0.150	4.3×†
Llama-3.1-70B	0.720	0.340	0.100	0.050	14.4×
MiniMax-M3	0.560	0.360	0.267	0.050	11.2×
Qwen3-4B	0.340	0.110	0.033	0.025	13.6×
Qwen3-8B	0.240	0.120	0.017	0.000	–

Table 53: Variety and degeneration over all 2,180 items (n: completions that `music21` parses into at least four notes, a looser test than the validation gate), with the IrishMAN reference sample of Table 56. Distinctness uses the first eight bars.

Source	n	PC ent.	Dist. bars	Dist. 4-gr.	≤3 pitches	Same bars	Rep. notes
IrishMAN corpus	299	2.63	0.828	0.769	0.0%	0.0%	0.7%
Qwen3-4B	1053	2.06	0.449	0.503	28.9%	5.4%	8.5%
Qwen3-8B	891	1.97	0.457	0.393	24.2%	18.0%	40.6%
Qwen3-32B	2076	2.64	0.775	0.752	0.8%	0.1%	5.0%
Llama-3.1-70B	2093	2.42	0.610	0.607	3.0%	0.2%	1.1%
MiniMax-M3	2143	2.29	0.725	0.794	9.2%	4.4%	12.7%
ChatMusician-7B	1710	1.83	0.599	0.551	31.2%	12.1%	32.8%
SFT	2180	1.87	0.389	0.390	23.0%	7.7%	32.0%
RFT	2172	2.07	0.472	0.521	31.7%	6.6%	11.2%
Direct GRPO ($R = A$)	2180	2.14	0.389	0.444	22.9%	7.8%	1.4%
MusicRLVR (SFT init.)	2180	1.62	0.359	0.342	35.1%	11.4%	40.7%
MusicRLVR (Direct)	2178	2.15	0.381	0.434	22.9%	6.5%	1.2%
MusicRLVR (Direct, Qwen3-8B)	2180	2.02	0.508	0.494	28.1%	7.0%	52.0%

Table 54: The same statistics on the 332 items without a form constraint that both Qwen3-4B and Direct MusicRLVR render to at least eight bars; every model is scored on those items.

Model	n	PC ent.	Dist. bars	Dist. 4-gr.	≤3 pitches	Same bars	Rep. notes
Qwen3-4B	332	2.02	0.530	0.562	30.1%	4.2%	12.3%
SFT	332	1.82	0.372	0.366	24.4%	9.3%	30.1%
Direct GRPO ($R = A$)	332	2.10	0.406	0.430	31.6%	16.9%	0.6%
MusicRLVR (Direct)	332	2.17	0.410	0.442	25.0%	7.5%	2.1%
MusicRLVR (Direct, Qwen3-8B)	332	2.23	0.627	0.634	16.3%	3.9%	61.1%
Llama-3.1-70B	322	2.40	0.665	0.631	4.7%	0.3%	2.2%

Table 58: Diagnostic study: Output length in the single-run mixed-suite diagnostic ($n = 300$ problems). Only completions that pass the historical parse gate are included, so the parsed count differs by model; the IrishMAN reference (299 tunes, 19.9 bars) is in Table 56.

Source	Parsed n	Mean bars	Mean notes
Qwen3-4B (zero-shot)	154	29.3	78.4
Qwen3-32B (zero-shot)	292	18.4	58.1
Llama-3.1-70B (zero-shot)	291	21.2	66.1
MiniMax-M3 (zero-shot)	173	17.9	37.0
SFT	300	39.2	167.5
MusicRLVR	300	28.7	130.3

D.25 Additional Training Curves

In Figure 9, we give the trainer-log quantities behind the diagnostics of Figure 7; the numbers behind it are as follows. Final KL to the reference falls $0.047 \to 0.024 \to 0.010$ as β rises $0.01 \to 0.04 \to 0.1$ (the $\beta = 0$ run logs no KL term), and yet the observed mixed rates lie in the narrow range 0.600–0.663 (Table 47): the KL term therefore tracks β as designed, while Mixed compliance is flat across the range. Along learning rate, final KL 0.006, 0.024, 0.049, 0.157 and mean group reward 0.713, 0.832, 0.893, 0.936 over 5×10^{-7} to 5×10^{-6} run in the same order as the mixed suite.

D.26 Additional Diagnostics

In Figure 10, we show two further per-item diagnostics: which families fail together, and when each item is first solved during training.

E Error and Structure Analysis

In this section, we analyse which constraints bind on each split and how form credit behaves across reward variants and form types.

E.1 Which Constraint Binds, by Split

In Figure 11, we give the outcome composition of each split, and in Table 60, we add the most frequent failing families, using the diagnostic reference evaluation. Figure 12 breaks the same evaluation down by requested form and by how many models solve each item.

Most unsatisfied items miss one family. The fraction missing multiple families is 0.040 on Seen and 0.105 on Unseen, rising to 0.206 on High-k. The most frequent failing family is `final` on Unseen and High-k. `structure` is among the three most frequent failures on Seen and Unseen; `leap` appears among the top three only on Single.

E.2 Structure Reward Variants

In Table 61, we compare the fixed default checkpoint with the equal-weight control and the symmetric-credit study. The default and equal-weight rows each use one run; symmetric credit reports the mean and sample SD of its two runs. The default form term weights repetition 0.7 and contrast 0.3; the following controls examine that choice.

Symmetric form credit. The contrast counterpart of Equation 3 is

$$\bar{d} = \frac{1}{|D|M} \sum_{(a,b)\in D} \sum_{m=1}^{M} \mathbb{I}\big[u_a[m] \neq u_b[m]\big],$$

Table 59: Diagnostic study: Fraction of GRPO groups with zero reward variance (duplicate step records from restarts removed), averaged over the first and last quarter of training and over all 300 steps, with the mean within-group reward standard deviation alongside.

	Zero-variance groups			Reward std	
Run	first 25%	last 25%	all	first 25%	last 25%
Binary form credit, original form share (21%)	0.116	0.307	0.221	0.301	0.214
Graded form credit, 33% form share (final)	0.133	0.322	0.254	0.296	0.210

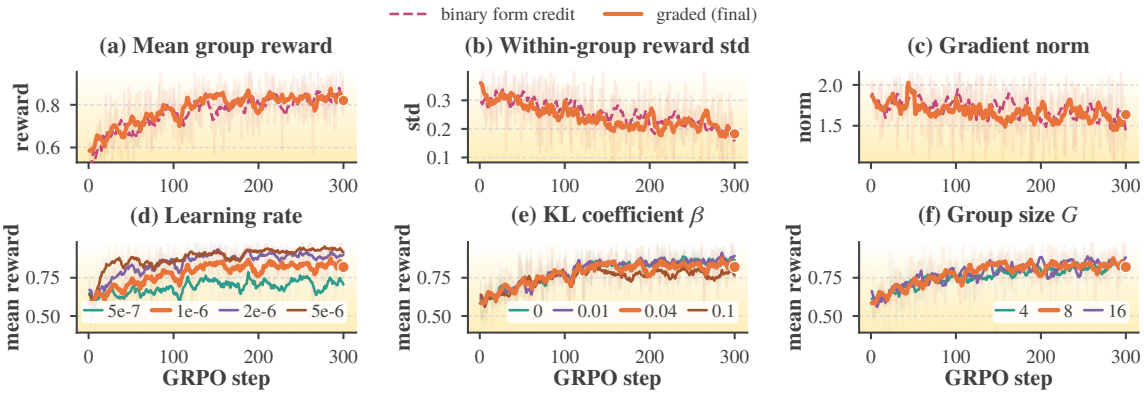


Figure 9: Diagnostic study: **trainer-log quantities behind Figure 7. (a–c)** Mean group reward, within-group reward SD and gradient norm, binary form credit against the final graded run. **(d–f)** Mean group reward for every hyper-parameter setting of Figure 7(b) on one shared scale; the thick blue curve is the default run (lr $= 1\text{e}{-6}$, $\beta = 0.04$, $G = 8$). Curves are smoothed over 9 steps; each axis spans the 2nd–98th percentile of its raw trace, so a few steps clip.

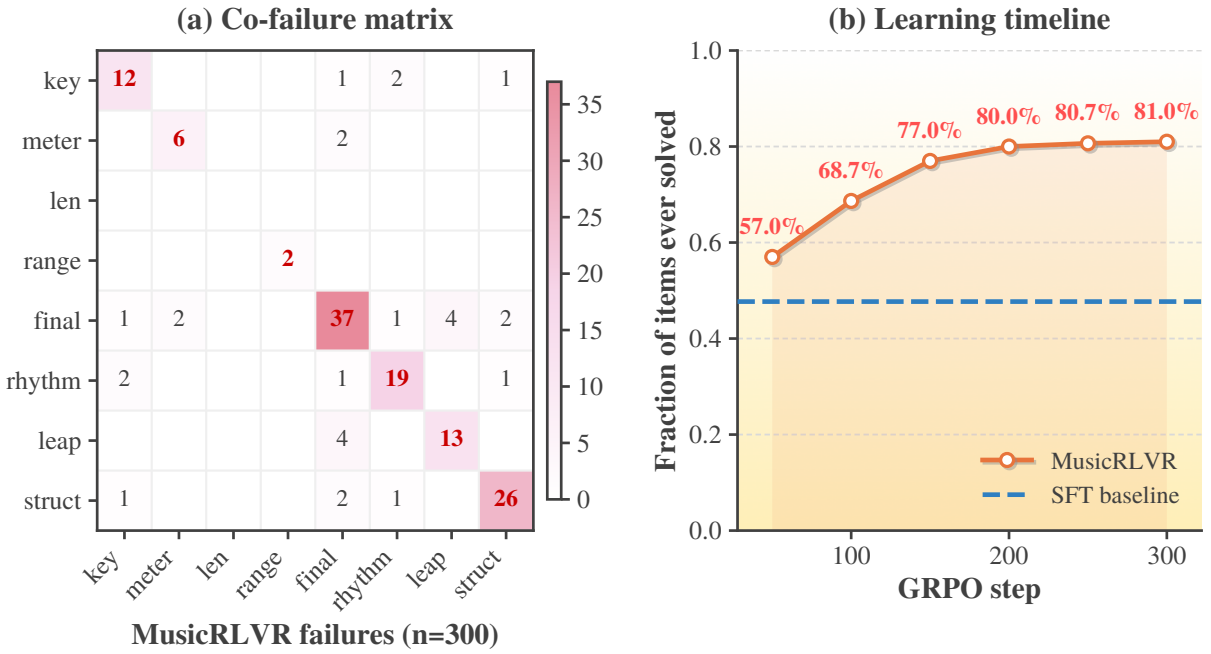


Figure 10: Diagnostic study: per-item diagnostics on the mixed suite ($n = 300$). **(a)** Co-failure matrix of MusicRLVR: diagonal entries are per-family failure counts, off-diagonal entries co-failures, which are rare: `key` shares at most two with any family. **(b)** CDF of the step at which each item is first solved: 57% by step 50, 81% at least once by step 300.

and the symmetric term is

$$s^{\text{sym}} = \begin{cases} 1, & v_{\texttt{structure}} = 1, \\ 0, & u_1 = u_2 = u_3 = u_4, \\ 0.5\,\bar{e} + 0.5\,\bar{d}, & \text{else.} \end{cases}$$

The outer reward mixture is unchanged. Across two runs, contrast satisfaction is 0.233 ± 0.024 and the all-identical fraction is 0.762 ± 0.018. The control jointly changes the weights, the contrast factor and the credit for all-identical outputs. Under the default term, an all-identical output earns $s_{\texttt{structure}} = 0.7$ but has $A = 0$, because it fails contrast. Appendix D.18 separately varies the outer mixing weight.

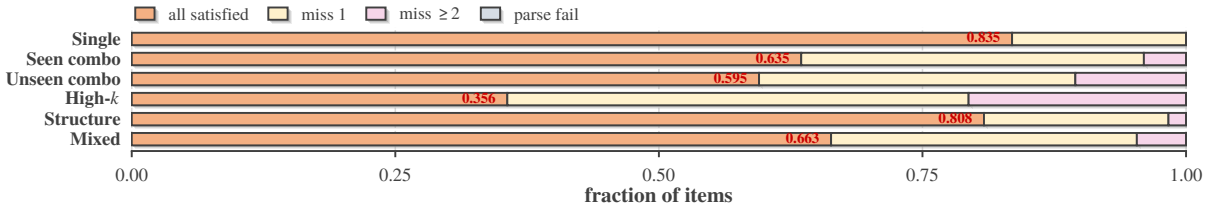


Figure 11: Diagnostic study: **What an unsatisfied item is missing.** Outcome composition for the fixed checkpoint; printed all-satisfied rates match Table 44. Parse failures are zero on all six original splits.

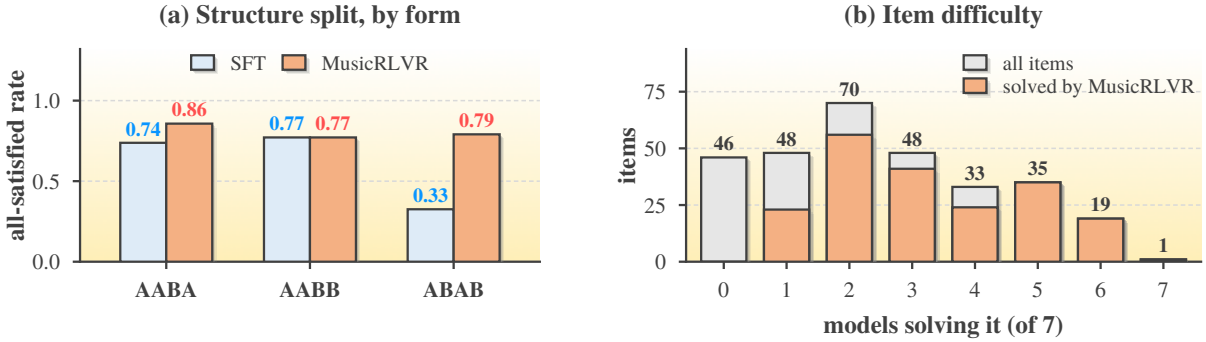


Figure 12: Diagnostic study: **Structure and item-level diagnostics. (a)** Satisfaction by requested form for the fixed checkpoint. **(b)** Item counts by the number of seven evaluated models solving them under the historical A.

Table 60: Diagnostic study: which constraints fail, by split. "Not all satisfied" is one minus the all-satisfied rate; the next column counts items failing two or more families at once; the last lists the three most frequent failing families as fractions of the split's items (an item failing two is counted under both). No completion fails the parse gate on any split.

Split	Not all satisfied	Failing ≥ 2 families	Most frequent failing families
Single	0.165	0.000	`leap` (0.085), `final` (0.025), `key` (0.020)
Seen	0.365	0.040	`structure` (0.125), `final` (0.120), `rhythm` (0.065)
Unseen	0.405	0.105	`final` (0.160), `key` (0.150), `structure` (0.090)
High-k	0.644	0.206	`final` (0.237), `rhythm` (0.181), `key` (0.156)
Structure	0.192	0.017	`structure` (0.133), `key` (0.042), `rhythm` (0.033)

Reweighting repetition and contrast. The single-variable version moves only the weights, to $0.5/0.5$: an all-identical output still banks the repetition half, the binary contrast factor is kept, and the rest of the reward and the training setup are unchanged. Two things follow.

(1) In the single-run diagnostic comparison, the structure split moves $0.808 \rightarrow 0.675$ and all-identical outputs $15 \rightarrow 32$ of 120, the same direction as the joint control (Table 61).

(2) The equal-weight control scores 0.637 on Mixed, compared with 0.663 for the default.

Appendix D.18 extends this comparison to the outer reward mixture. In the single-run diagnostic study, the two tested λ endpoints yield larger increases in all-identical outputs than the equal-weight form control. The default achieves the highest Structure score and fewest all-identical outputs among these tested weight settings. In the Direct configuration (Table 6), the default again gives the highest Structure, 0.967 against 0.942 for both variants, and the highest rate over all items, 0.580 against 0.515 (symmetric) and 0.544 (equal weights).

E.3 Structure Results by Form Type

In Table 62, we decompose performance on the three form types of the structure split, which come in unequal numbers: AABA ($n = 42$), AABB ($n = 35$) and ABAB ($n = 43$).

Table 62: Diagnostic study: Structure results by form for the fixed MusicRLVR and SFT checkpoints. Contrast and all-identical are fractions of items of that form.

Model	Form	All-sat.	Contrast	All-identical
SFT	AABA	0.738	0.810	0.190
SFT	AABB	0.771	0.857	0.143
SFT	ABAB	0.326	0.395	0.605
MusicRLVR	AABA	0.857	0.905	0.071
MusicRLVR	AABB	0.771	0.829	0.171
MusicRLVR	ABAB	0.791	0.860	0.140

ABAB is the form supervision handles worst, and by a wide margin: SFT reaches 0.771 on AABB but 0.326 on ABAB, with $26/43$ of its ABAB outputs written as four identical units. All three

Table 61: Diagnostic study: structure decomposition. "Runs" is the number of training runs: default and equal-weight use one each, symmetric credit reports mean ± sample SD over two. All-identical is a fraction; "all requested" additionally requires the companion constraint, present in 66 of 120 items.

Reward	Runs	Repetition	Contrast	All requested	All-identical
MusicRLVR	1	0.992	0.867	0.808	0.125
Equal weights	1	1.000	0.733	0.675	0.267
Symmetric credit	2	0.996 ± 0.006	0.233 ± 0.024	0.208 ± 0.012	0.762 ± 0.018

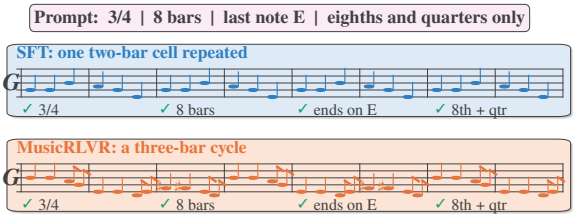


Figure 13: One High-k item, diagnostic study (Appendix D.14): both outputs satisfy all four constraints.

forms require distinct A and B units, so the gap tracks the arrangement rather than the distinctness requirement: ABAB is the only one that interleaves the two units instead of repeating each in place.

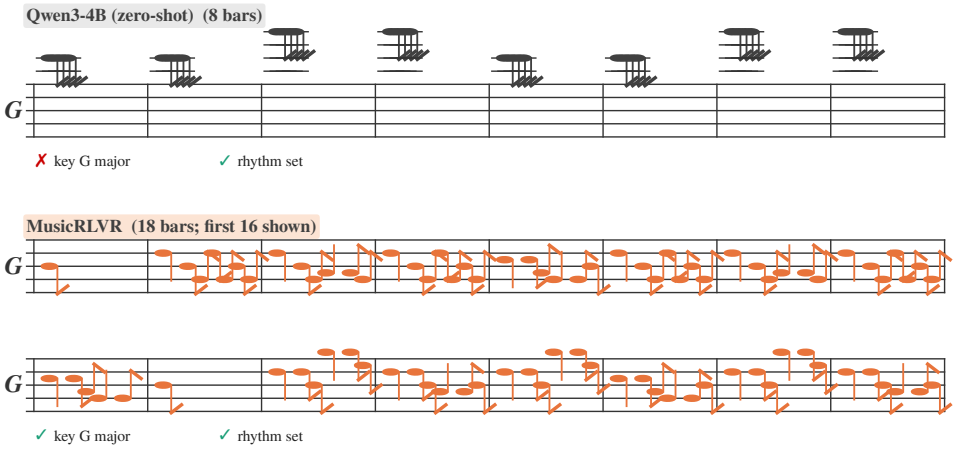
At the fixed checkpoint, ABAB has the largest improvement over SFT among the three forms: its all-satisfied rate rises from 0.326 to 0.791, while the all-identical fraction changes from 0.605 to 0.140. Table 62 reports the rate for every form.

F Qualitative Examples

In this section, we present qualitative examples. Figure 13 shows two outputs that satisfy the same four constraints with different repeating patterns. Every example below is a benchmark item chosen by a fixed rule (the first item in file order with the stated constraint families and pass/fail pattern), its completions are read verbatim from the frozen evaluation files, and the notation is drawn from `music21`'s parse of the output: pitch, duration and bar lines. The key and meter checks additionally read the header-derived objects, and the key check analyses the whole score. Marks under each staff are the verifiers' decisions on that output. Outputs rejected at the gate are shown as a short excerpt together with the gate's reason. Examples 1–6 compare the zero-shot base model with MusicRLVR (diagnostic run), and Example 2 shows SFT as well; Examples 7–8 compare MusicRLVR with the larger zero-shot models on the same items.

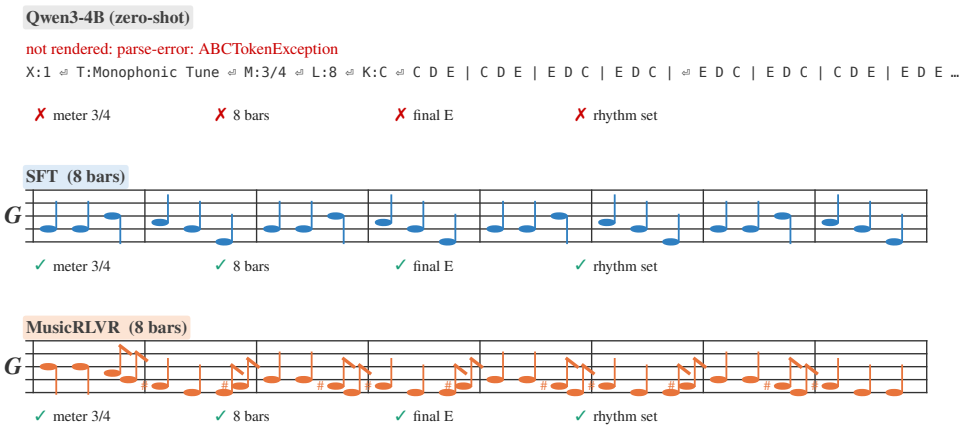
Examples 9–14 use the matched Direct MusicRLVR of Table 2 and one rule: the first item in file order, among those with the stated families, that Direct MusicRLVR satisfies while Qwen3-32B, Llama-3.1-70B and MiniMax-M3 all fail. Such items are common. Over the 2,180 benchmark items, Direct MusicRLVR satisfies 609 that all three of these models fail, whereas all three satisfy only 5 items that Direct MusicRLVR fails. The six examples cover four constraints on Mixed, two held-out pairs on Dense (with the Qwen3-8B policy alongside), a held-out pair with a form on Unseen, the interleaved ABAB form, and two Edge items: one with out-of-range meter ($5/4$), and one combining a rhythm restriction with ABAB structure.

Example 1: unseen-combination split ($k = 2$, held-out pair `key–rhythm`).
Item `uc-0022`. **Constraints:** (1) the tune must be in G major; (2) use only sixteenth, eighth, and quarter notes (no dotted notes, no other durations, no rests)



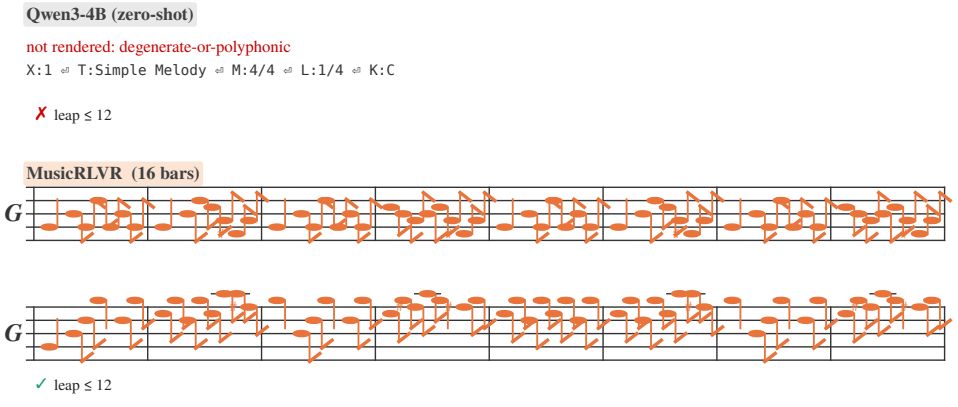
Qwen3-4B (zero-shot) fails `key`; MusicRLVR satisfies every constraint.

Example 2: high-k split ($k = 4$).
Item `hk-0004`. **Constraints:** (1) the meter must be 3/4, and every bar must contain exactly the right total duration; (2) the tune must be exactly 8 bars long (no pickup bar); (3) the last note of the tune must be E (any octave); (4) use only eighth and quarter notes (no dotted notes, no other durations, no rests)



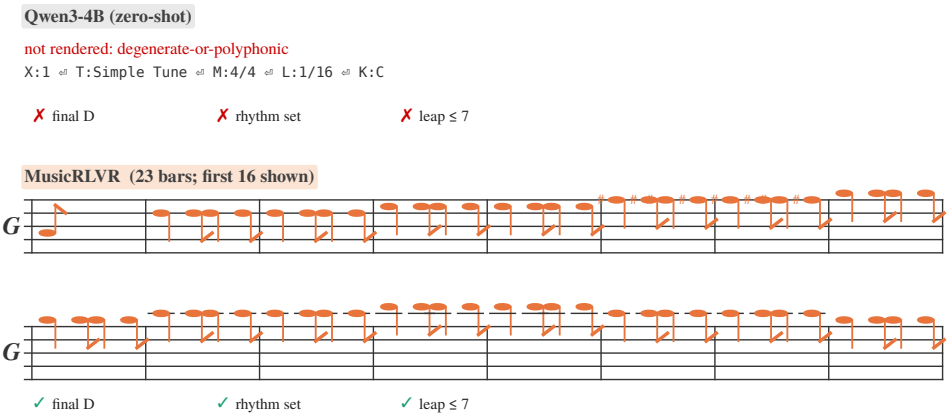
Qwen3-4B (zero-shot) fails at the parse gate (parse-error: ABCTokenException); SFT satisfies every constraint; MusicRLVR satisfies every constraint.

Example 3: single-constraint split ($k = 1$).
Item `sg-0000`. **Constraints:** (1) melodic motion must be smooth: no melodic leap larger than an octave



Qwen3-4B (zero-shot) fails at the parse gate (degenerate-or-polyphonic); MusicRLVR satisfies every constraint.

Example 4: seen-combination split ($k = 3$).
Item `sc-0005`. **Constraints:** (1) the last note of the tune must be D (any octave); (2) use only sixteenth, eighth, and quarter notes (no dotted notes, no other durations, no rests); (3) melodic motion must be smooth: no melodic leap larger than a perfect fifth



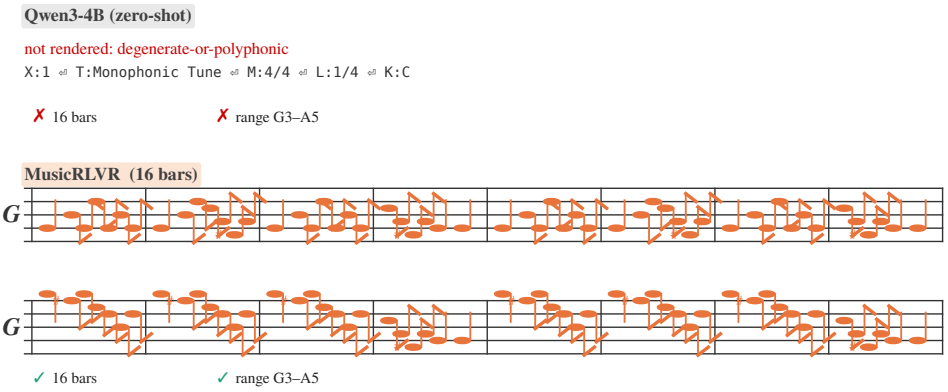
Qwen3-4B (zero-shot) fails at the parse gate (degenerate-or-polyphonic); MusicRLVR satisfies every constraint.

Example 5: seen-combination split ($k = 2$).
Item `sc-0001`. **Constraints:** (1) use only sixteenth, eighth, and quarter notes (no dotted notes, no other durations, no rests); (2) the tune must be exactly 16 bars with an AABB form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ



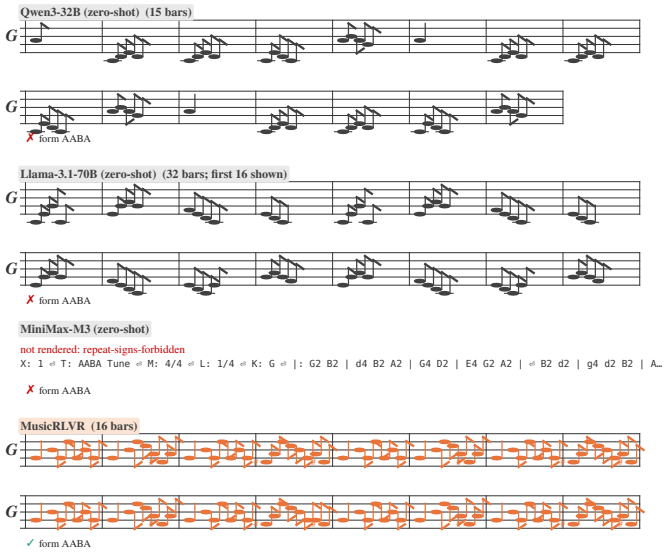
Qwen3-4B (zero-shot) fails `structure`; MusicRLVR satisfies every constraint.

Example 6: unseen-combination split ($k = 2$, held-out pair `length–range`).
Item `uc-0001`. **Constraints:** (1) the tune must be exactly 16 bars long (no pickup bar); (2) every note must lie between G3 and A5 inclusive



Qwen3-4B (zero-shot) fails at the parse gate (degenerate-or-polyphonic); MusicRLVR satisfies every constraint.

Example 7: structure split ($k = 1$, zero-shot models against MusicRLVR).
Item `st-0023`. **Constraints:** (1) the tune must be exactly 16 bars with an AABA form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ



Qwen3-32B (zero-shot) fails `structure`; Llama-3.1-70B (zero-shot) fails `structure`; MiniMax-M3 (zero-shot) fails at the parse gate (repeat-signs-forbidden); MusicRLVR satisfies every constraint.

Example 8: high-k split ($k = 4$, zero-shot models against MusicRLVR).
Item `hk-0278`. **Constraints:** (1) the tune must be in C major; (2) every note must lie between D4 and D6 inclusive; (3) the last note of the tune must be G (any octave); (4) melodic motion must be smooth: no melodic leap larger than a perfect fifth



Qwen3-32B (zero-shot) fails `range`, `final`; Llama-3.1-70B (zero-shot) fails `range`; MusicRLVR satisfies every constraint.

Example 9: Mixed split ($k = 4$).
Item `k4-0002`. **Constraints:** (1) the tune must be in A minor; (2) the meter must be 3/4, and every bar must contain exactly the right total duration; (3) the last note of the tune must be C (any octave); (4) melodic motion must be smooth: no melodic leap larger than a perfect fifth

Qwen3-32B (zero-shot) (8 bars)



✓ key A minor ✗ meter 3/4 ✓ final C ✗ leap ≤ 7

Llama-3.1-70B (zero-shot) (20 bars; first 8 shown)



✓ key A minor ✗ meter 3/4 ✓ final C ✗ leap ≤ 7

MiniMax-M3 (zero-shot) (32 bars; ABC excerpt)

ABC excerpt; fails meter and leap

```
X: 1 ⏎ T: Smooth A Minor Waltz ⏎ M: 3/4 ⏎ L: 1/4 ⏎ K: Am ⏎ A2 c2 | e2 d2 | c2 A2 | B2 c2 | ⏎ d2 e2 | c2…
```

✓ key A minor ✗ meter 3/4 ✓ final C ✗ leap ≤ 7

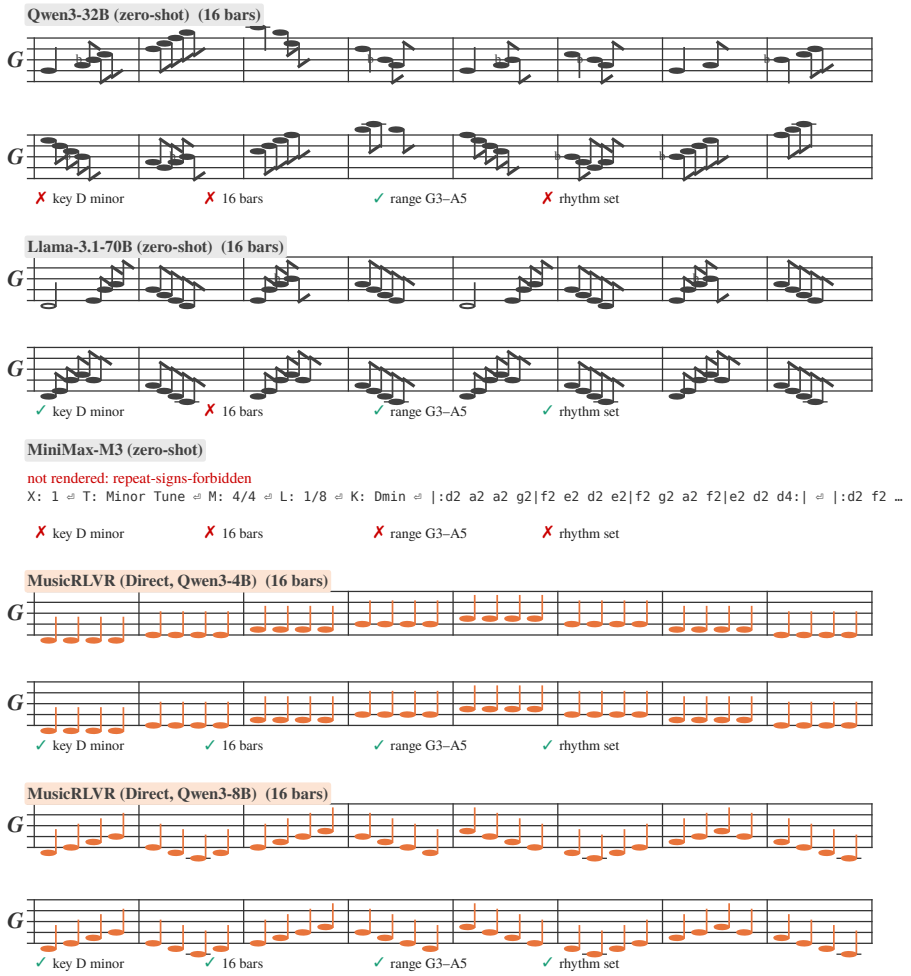
MusicRLVR (Direct, Qwen3-4B) (12 bars; first 8 shown)



✓ key A minor ✓ meter 3/4 ✓ final C ✓ leap ≤ 7

Qwen3-32B (zero-shot) fails `meter`, `leap`; Llama-3.1-70B (zero-shot) fails `meter`, `leap`; MiniMax-M3 (zero-shot) fails `meter`, `leap`; MusicRLVR (Direct, 4B) satisfies every constraint.

Example 10: Dense split ($k = 4$, two held-out pairs, MusicRLVR on both base models).
Item dn-0003. **Constraints:** (1) the tune must be in D minor; (2) the tune must be exactly 16 bars long (no pickup bar); (3) every note must lie between G3 and A5 inclusive; (4) use only eighth, quarter, and half notes (no dotted notes, no other durations, no rests)



Qwen3-32B (zero-shot) fails `key`, `length`, `rhythm`; Llama-3.1-70B (zero-shot) fails `length`; MiniMax-M3 (zero-shot) fails at the parse gate (repeat-signs-forbidden); MusicRLVR (Direct, 4B) satisfies every constraint; MusicRLVR (Direct, 8B) satisfies every constraint.

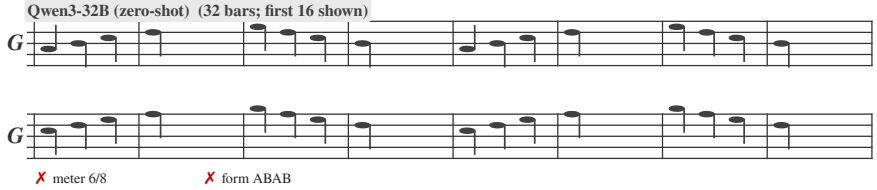
Example 11: unseen-combination split ($k = 3$, held-out pair `meter–leap` plus a form).
Item `uc-0006`. **Constraints:** (1) the meter must be 6/8, and every bar must contain exactly the right total duration; (2) melodic motion must be smooth: no melodic leap larger than an octave; (3) the tune must be exactly 16 bars with an ABAB form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ



Qwen3-32B (zero-shot) fails `meter`, `structure`; Llama-3.1-70B (zero-shot) fails `meter`, `structure`; MiniMax-M3 (zero-shot) fails at the parse gate (repeat-signs-forbidden); MusicRLVR (Direct, 4B) satisfies every constraint.

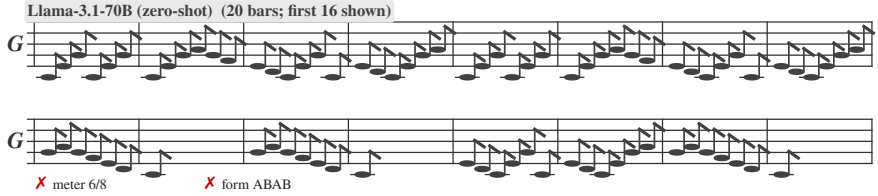
Example 12: structure split ($k = 2$, the interleaved form ABAB).
Item `st-0033`. **Constraints:** (1) the meter must be 6/8, and every bar must contain exactly the right total duration; (2) the tune must be exactly 16 bars with an ABAB form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ

Qwen3-32B (zero-shot) (32 bars; first 16 shown)



✗ meter 6/8 ✗ form ABAB

Llama-3.1-70B (zero-shot) (20 bars; first 16 shown)



✗ meter 6/8 ✗ form ABAB

MiniMax-M3 (zero-shot)

not rendered: repeat-signs-forbidden

```
X:1 ⏎ T:Monophonic Tune ⏎ M:6/8 ⏎ L:1/8 ⏎ K:G ⏎ |: D | G2 B d2 B | A2 F D2 F | G2 A B2 c | d2 c B2 A | ⏎ G2 B …
```

✗ meter 6/8 ✗ form ABAB

MusicRLVR (Direct, Qwen3-4B) (16 bars)

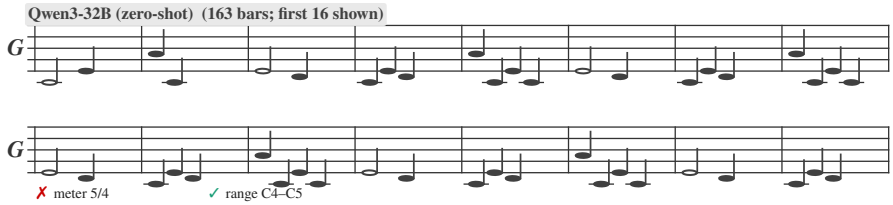


✓ meter 6/8 ✓ form ABAB

Qwen3-32B (zero-shot) fails `meter`, `structure`; Llama-3.1-70B (zero-shot) fails `meter`, `structure`; MiniMax-M3 (zero-shot) fails at the parse gate (repeat-signs-forbidden); MusicRLVR (Direct, 4B) satisfies every constraint.

Example 13: Edge split ($k = 2$, a $5/4$ meter, outside the RL prompt range).
Item `eg-0033`. **Constraints:** (1) the meter must be 5/4, and every bar must contain exactly the right total duration; (2) every note must lie between C4 and C5 inclusive

Qwen3-32B (zero-shot) (163 bars; first 16 shown)



✗ meter 5/4 ✓ range C4–C5

Llama-3.1-70B (zero-shot) (12 bars)



✗ meter 5/4 ✓ range C4–C5

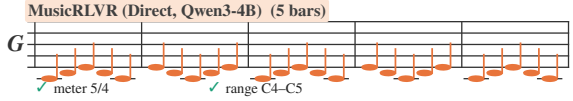
MiniMax-M3 (zero-shot)

not rendered: degenerate-or-polyphonic

```
X: 1 ⏎ T: Five-Four Tune ⏎ M: 5/4 ⏎ L: 1/4 ⏎ K: C
```

✗ meter 5/4 ✗ range C4–C5

MusicRLVR (Direct, Qwen3-4B) (5 bars)

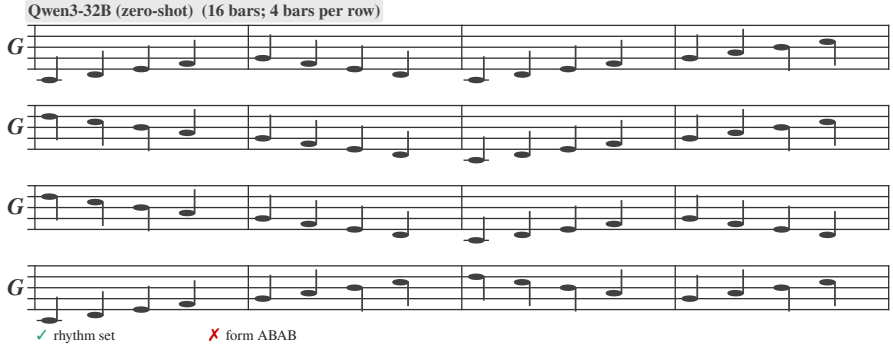


✓ meter 5/4 ✓ range C4–C5

Qwen3-32B (zero-shot) fails `meter`; Llama-3.1-70B (zero-shot) fails `meter`; MiniMax-M3 (zero-shot) fails at the parse gate (degenerate-or-polyphonic); MusicRLVR (Direct, 4B) satisfies every constraint.

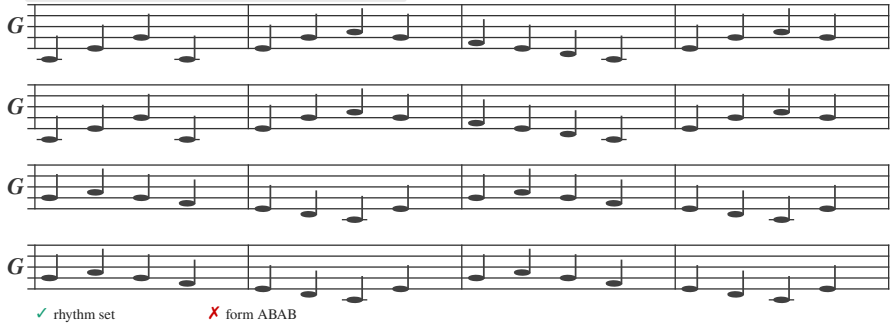
Example 14: Edge split ($k = 2$, rhythm restriction + ABAB structure).
Item `eg-0005`. **Constraints:** (1) use only quarter and half notes (no dotted notes, no other durations, no rests); (2) the tune must be exactly 16 bars with an ABAB form in 4-bar units: labeled units with the same letter must be note-for-note identical, and units with different letters must differ

Qwen3-32B (zero-shot) (16 bars; 4 bars per row)



✓ rhythm set ✗ form ABAB

Llama-3.1-70B (zero-shot) (16 bars; 4 bars per row)



✓ rhythm set ✗ form ABAB

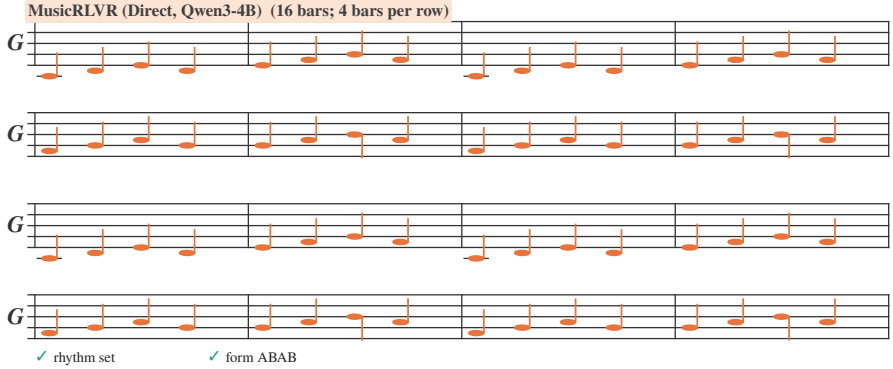
MiniMax-M3 (zero-shot)

not rendered: repeat-signs-forbidden

```
X: 1 ⏎ T: ABAB Tune ⏎ M: 4/4 ⏎ L: 1/4 ⏎ K: G ⏎ |: G2 B2 | d2 B2 | A2 G2 | E2 D2 | ⏎ G2 B2 | d2 B2 | A2 …
```

✗ rhythm set ✗ form ABAB

MusicRLVR (Direct, Qwen3-4B) (16 bars; 4 bars per row)



✓ rhythm set ✓ form ABAB

Qwen3-32B (zero-shot) fails `structure`; Llama-3.1-70B (zero-shot) fails `structure`; MiniMax-M3 (zero-shot) fails at the parse gate (repeat-signs-forbidden); MusicRLVR (Direct, 4B) satisfies every constraint.

G Broader Impact

In this section, we expand the ethics statement. The method makes music-generation models follow explicit specifications, which helps wherever a score must meet a specification: music education, game audio, accessibility tools. We do not foresee significant negative societal impacts specific to this work, which concerns short monophonic melodies in ABC notation under explicit constraints. The likeliest misreading is to take a high all-satisfied rate as a quality score; it certifies that the requested properties hold. All supervised targets derive from the publicly available IrishMAN corpus of traditional Irish music; the reinforcement prompts are generated synthetically. The benchmark and verifiers will be released to support reproducibility and further research.